CELL

Robert Hooke	discoverd cell in 1665
Leeuwenhoek	discoverd the free Living cell in pond water. 1st
Robert Brown	Discovery of Nucleus
purkinje	term 'protoplasm' for fluid sub. of the cell
Virchow	"All cells arise from pre-existing cells"
Schleiden & Schwann	The cell Theory →

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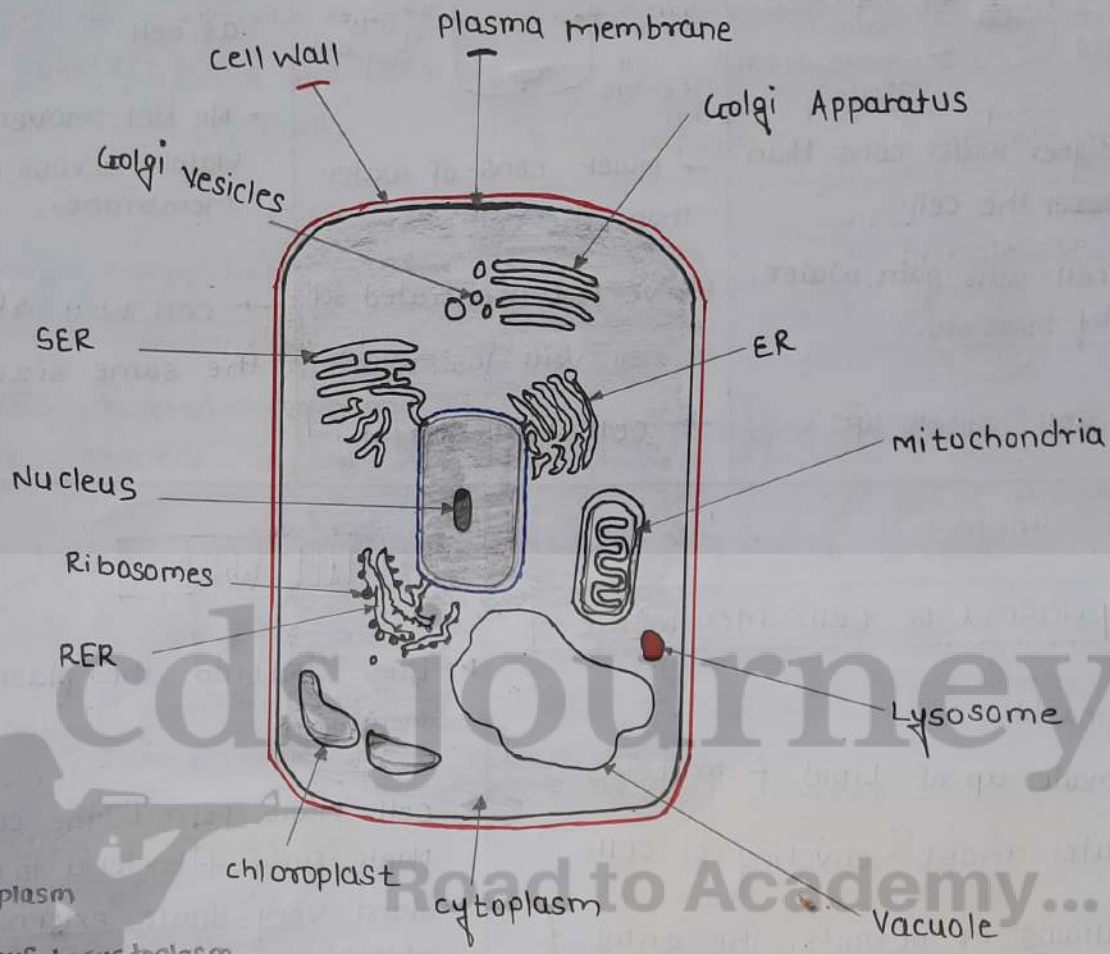
Unicellular	multicellular
- made of one cell	- more than one cell
- single cell performs all life functions.	- division of labour
	- specialized role
- Ex: Amoeba, Bacteria chlamydomonas, paramesium * binary fission	- Ex: - plant - Animal - fungi

Division of labour is also seen within a single cell.

- Bigest cell - ostrich egg
- longest cell - Neurons
- smallest cell - Mycoplasma.



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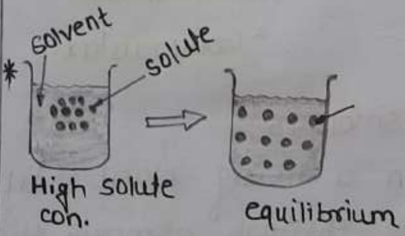
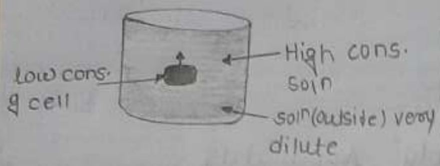
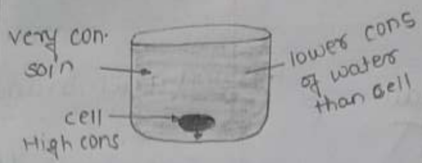
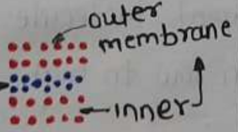
*protoplasm
↓
Nucleus + cystoplasm

Diffusion	Osmosis.
* movement of a substance from a region of H·C to a L·C.	* The movement of water from H·C to l·C through selectively / semi permeable membrale.
* solute molecules moves from high to low con.	* solvent molecule move from low to high solute co.
* 	* Unicellular freshwater organism gain H_2O through osmosis
	* Absorption of water by plant roots.

Hypotonic	Hypertonic	isotonic
		→ same water cons. as cell.
→ Higher water cons. than ~~lower~~ the cell.	→ lower cons. of water than the cell.	→ No net movement of water. across cell. membrane.
→ cell will gain water by osmosis.	→ very concentrated solⁿ → cell will lose water	→ cell will stay the same size.
→ cell swell up	→ cell will shrink	

Plasma or Cell Membrane.

- flexible
- made up of Lipid + proteins
- outer most covering of cell.
- allows or permits the entry + exits of some materials. in + out of cell
- called sectively permeable membr.
- some substances like carbon dioxid or O_2 move across cell membrane by a process of diffusion. (HC → LC)
- hydrophobic region



The flexibility of the cell membrane also enables the cell to engulf in food + other materials from its external environment. called endocytosis. → Amoba acquires its food through...

CELL WALL

- Lies outside the plasma membrane.
- cell walls permit the cells of plants, Fungi + bacteria to withstand very dilute external media without bursting.
- cell wall :
  - Plants → cellulose
  - Fungi → chitin
  - Bacteria → peptidoglycan
  - Diatom
  - Mushroom
  - paramecium
  - Gonyaculax.
- Plasmolysis.

When a living plant cell loses water through osmosis there is shrinkage or contraction of the contents of the cell away from the cell wall.

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- provides structural strength to the plants.
- Help protect the cell in hypotonic external medium.

NUCLEUS

- double layered covering called Nuclear membrane.
- N.M has pores which allow the transfer of material.

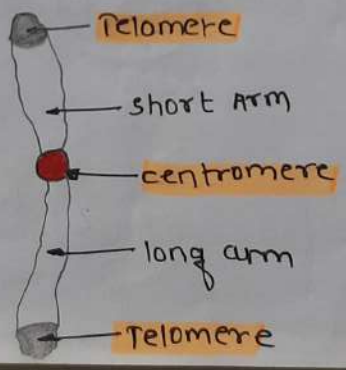
\# Nucleus
↓
contains chromosomes

→ contain inf. for inheritance of characters from parents to next generation in the form of DNA molecules.

→ composed of DNA & proteins
↓
functional segment of DNA are known as genes.

→ Rod shaped structure


chromatids



* In a cell which is not dividing this DNA is present as part of cromatin material.

* choromosomes	chromatin
	
1] tightly packaged DNA.	i) unwond DNA
2] Found only during cell division	ii) found throughout Interphase.
3] DNA is not being used for macromolecule synthesis	iii) DNA is being used macromolecule synthesis.

→ Chromatin fibres
- present inside the Nucleus.
- made up of histone protein & DNA
- thickens to form chromosomes during the process of the cell division.

Heter
(tig
Found
gene

Fun^n: re
expression

tin
ked)

ns-
cription of DNA to mRNA.

- Some organisms like bacteria Nuclear region of the cell may be poorly defined due to absence of Nuclear membrane.
- undefined Nuclear region
↓
contain → Nucleic Acid. (Nucloid)

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→ lack of a Nuclear membrane are called prokaryotes.

Eukaryotes	Prokaryotes
► size - large	* small
► well Defined Nuclear.	* Nucl. absent
	* Nucleoid present
► chromosome: ~~sin~~ more than one	* single
mem. bound cell org. absent.	
► Ex: ~~Bacteria~~ • plant • animals • Fungi • protists	Ex: • Bacteria • mycoplasma • blue green alagae • Archaea
	* membrane enclosed organells

common to pro. & many euk.

↓

- * Cell Wall
- * Ribosome
- * Cell membrane

chromatin material is present in eukaryotes only.

Nucleols - ribosome syntheses.

Viruses - lack of membranes.

eukaryotic ~~prot~~ chromatin is composed 7.

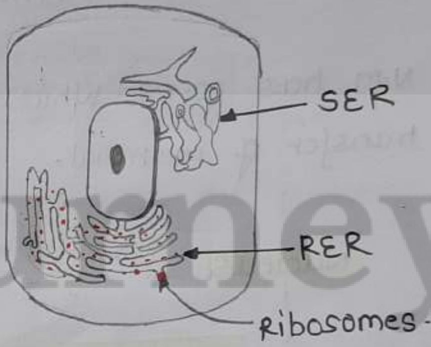
- protein & DNA.

ER

→ Fun: transport of material betn cytoplasm & the Nucleus.

→ similar in structure to the plasma membrane

→ large Network of membrane bound tubes & and sheets.



ER

→ RER → Rough surface due to presence of ribosomes → protein synthesis → send through ER to Nucleous.

→ SER → help in Lipid formation & fat molecules. → role in detoxifying many poisons & drugs. → # In vertabrate → Liver → detoxifying poison

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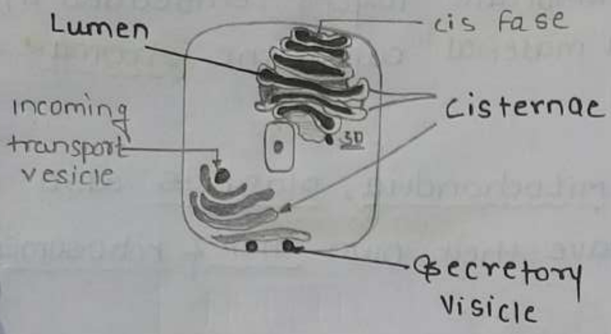
GOLGI APPARATUS



membrane - bound vesicles (flattened sacs) arranged approximately parallel to each other in stacks called cisterns.

Fun^n:
- storage
- modification
- packaging

complex sugar made simple sugar.

→ involved in the formation of lysosomes.

Absent in mychoplasm

LYSOSOMES

→ Have digestive enzymes.

► Waste disposal system of the cell.

► laysosomes are membrane-bound sacs filled with digestive enzymes.

► These enzymes are made by RER.

► suicide bags of a cell digest their own cell.

► help to keep the cell clean. ~~by~~

► break complex substances into simpler substances.

VACUOLES

• storage sacs for solid or liquid contents.

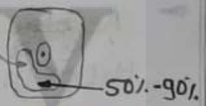
Vacuoles

Animals

small sized

plant

- very large sized.



provide turgidity + rigidity. to plant cell.

plant cell stored in vacules

- Amino acid
- sugar
- organic Acids
- some proteins

In Amoeba, vacuoles have role in Nutrition.

MITOCHONDRIA

— power house of cell

— two membrane covering

outer (porous)

inner (deeply folded)

— energy released by mitochondria in the form of ATP.

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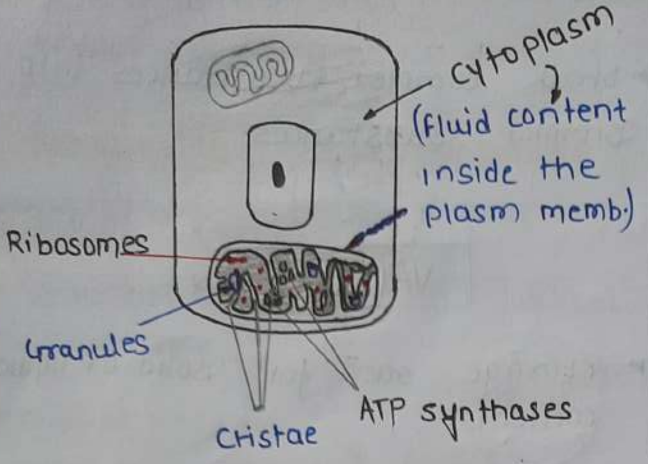


- Has its own DNA & Ribosomes
- ATP - energy currency of the cell
- able to make theire won protein.
- Aerobic respiration takes place in the mitochondiria

PLASTIDS

- only in plant cell.
- plastids
  - Chromoplasts (coloured)
    - containing the pigment chlorophyll
    - are known chloroplasts.
  - Leucoplasts (colorless)
    - Stored:
      - starch
      - oils
      - protein granules

● chloroplasts:
Imp for photo-syntesis in plant

● chloroplasts:
caintain verious yellow & orange pigment.

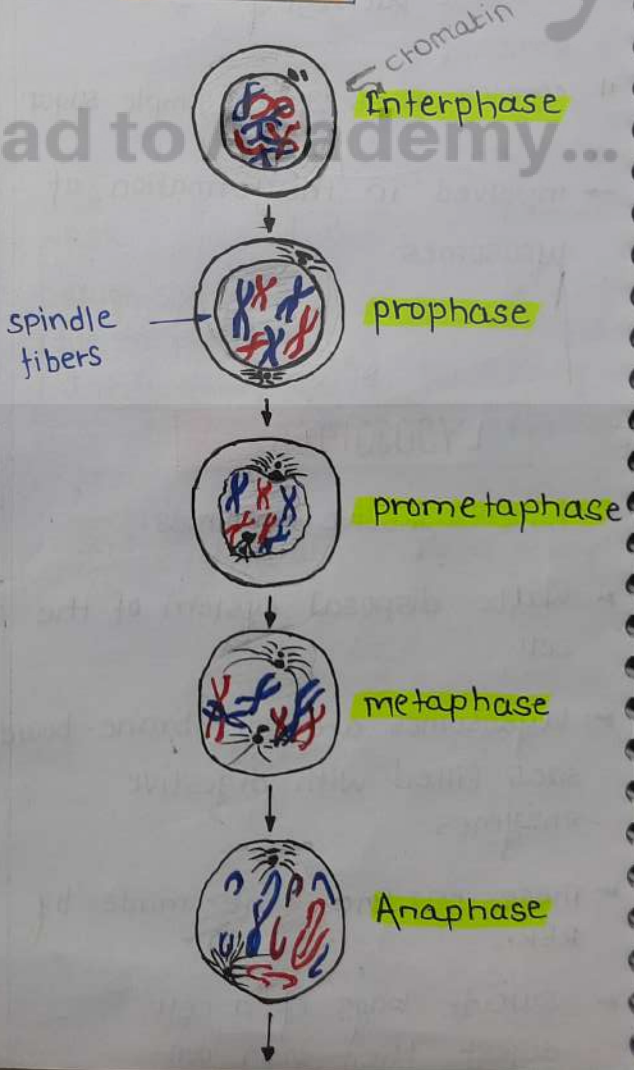
- The internal organisation of the chloroplast consists of Numerous membrane layers embedded in a material called the stroma.

→ mitochondria, plastids also have their own DNA & ribosomes.

CELL DIVISION

→ the process by which new cells are made is called cell devision.

Mitosis



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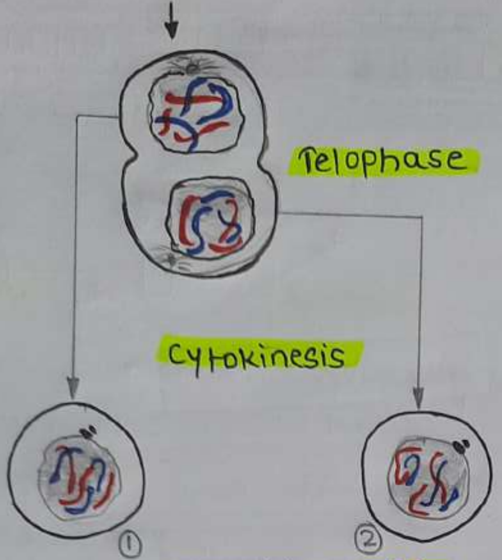
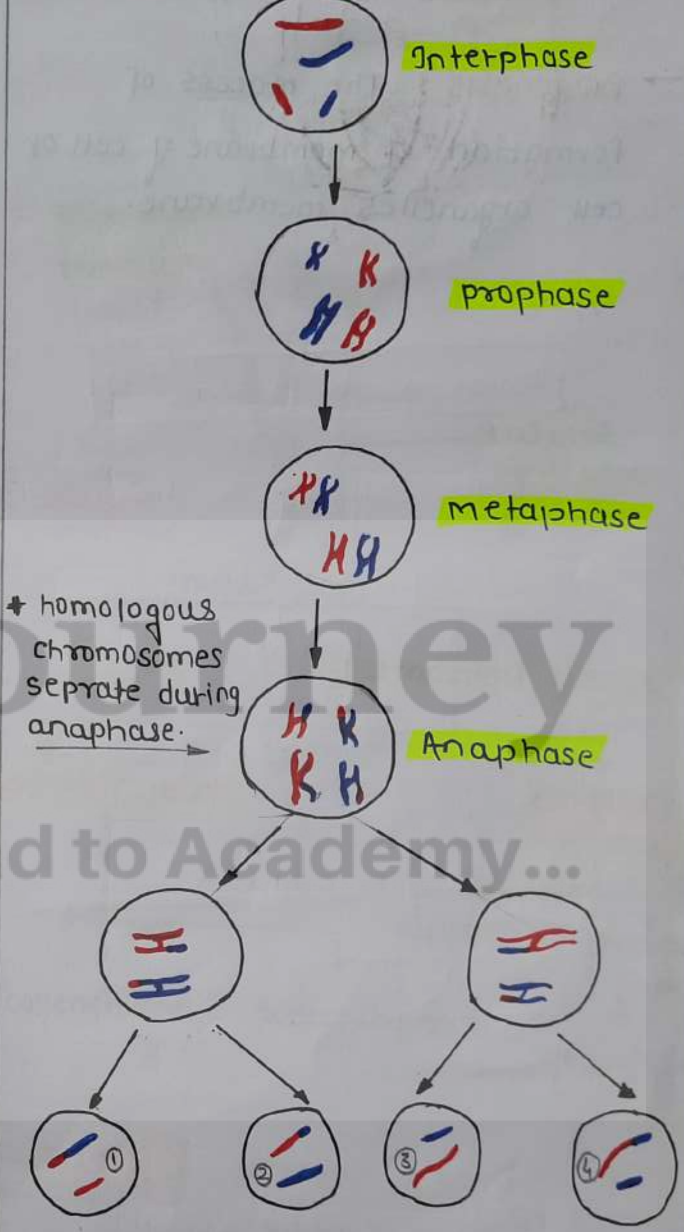


2 daughter cells

- each cell called mother cell divides to form two identical daughter cell.
- daugher cells have same no. of chromosomes. as mother cell.
- helps in growth & repair of tissues in organisms.

MEIOSIS



4 daughter cells

→ involves two consecutive division

→ When cell divides by meiosis it produces four New cells instead of just two.

→ New cell only have half the no. of chromosomes than that of mother cells.

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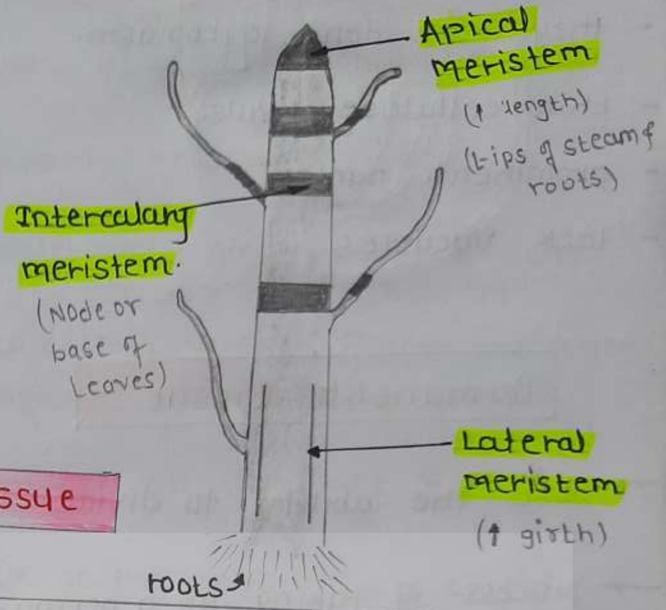
TISSUES

- Tissues
  - Plant
    - large quantity of supportive tissue
    - S.t generally dead cell.
  - Animals
    - most are living



Plant tissue

- Meristematic tissue
  - Apical
  - Lateral (cambian)
  - Intercalary
- permanent tissue
  - simple/Ground
    - parenchyma
    - collenchyma
    - Scelrenchyma
  - complex/vascular
    - xylem
    - Phloem

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Meristematic t.

Apical

- present at growing tips of stems & roots
- Increase the length of the stem & roots.
-

Lateral (cambian)

- increase girth of the stem or root

Intercalary

- some plants is located near the Node or base of Leaves.

* cell of maristematic tissue are very active.

- they have dense cytoplasm.
- thin cellulose walls.
- prominent nuclei.
- lack vacuoles.

Permanent tissue

→ lose the ability to divide.

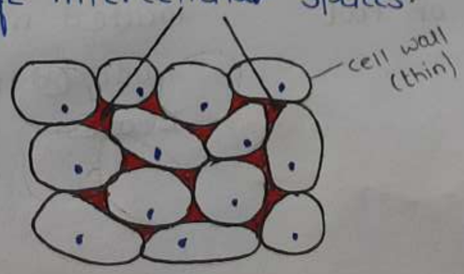
→ process of taking up a permanet shape, size & a fun^{n}.

→ all made of one type of cell.

1) Simple permanent tissue

Parenchyma — stores food.

- most common simple p. tissue.
- consists of relatively unspecialised cells with thin cell walls.
- they are living cells
- Loosely packed
- Large intercellular spaces.

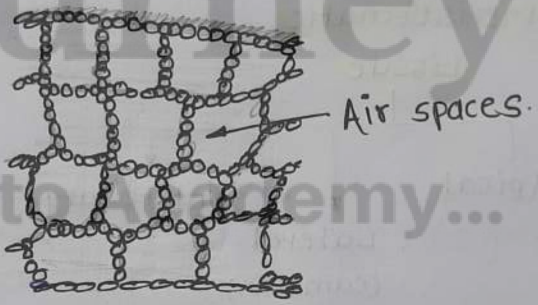


Chlorenchyma

- contain chlorophyll & performs photosynthesis

Arenchyma

- In aquatic plants.
- large air cavities in paranchyma to help them float.
- also write parenchyma.



collenchyma

- mechanical support & flexibility
- Live cells
- cells thickened at the corners
- Little intercellular space.
- Find : leaf stalks below the epidermis.
- allow to bending (Flexibility).

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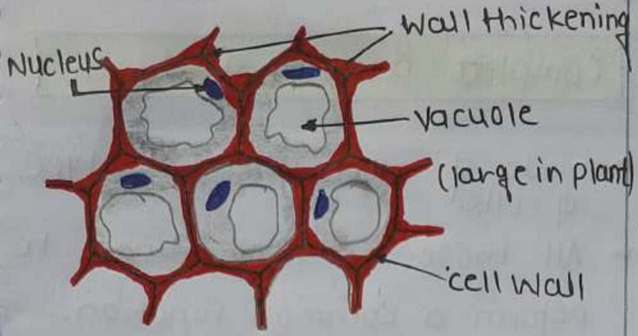

Sclerenchyma

- provides hardness & stiff
- Dead cells.
- Long Narrow cells
- Walls thickened due to lignin
- No intercellular spaces
- husk of coconut made by Sclerenchymatous.
- present : - in stems, around
  - vascular bundles
  - in the veins of leaves
- provides strength to the plant parts.

meriste.	permanent
living - cells ® small	both L+D - cells ® large
- vacuoles absent	- present
- Intercellular space less - absent	- Int. space ® present.
- cell wall - thin	- cell wall - thick
- Nucleus is prominent	- Nucleus is less visible.

Epidermis

- outermost single layer of cells.
- epidermis thicker since protection against water loss is critical.
- protect all the part of plants.
- E.C on the arerial parts of the plant often secrete a waxy, water resistant layer on their outer surface.

↓

this aids in protection against loss of water
- mechanical injury
- invasion by parasitic fungi.

- continuous layer without intercellular space.
- most ® relatively flat.
- outer & side wall → thicker

Stomata

- Small pores here & there in the epidermis of the leaf.

↓

these pores are stomata.

- enclosed by two kidney shaped cells called guard cells.

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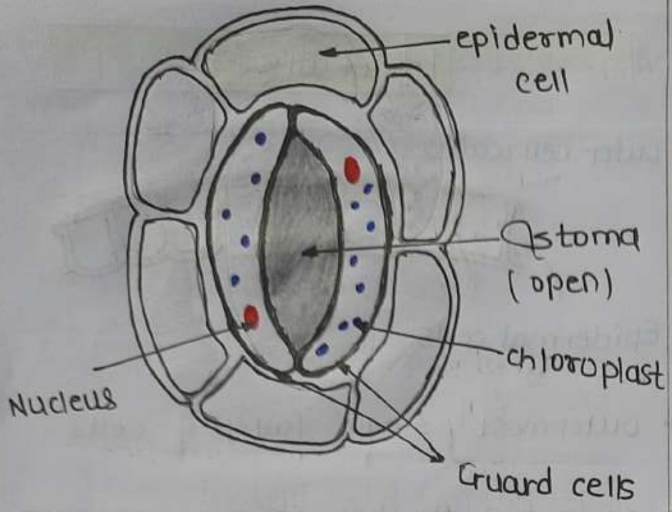


- Transpiration also takes place through stomata.

Transpiration: loss of H_2O in the form of water vapour.

- Epidermal cells of the roots whose function is water absorption.

→ cells of cork are dead and no intercellular space.

↓

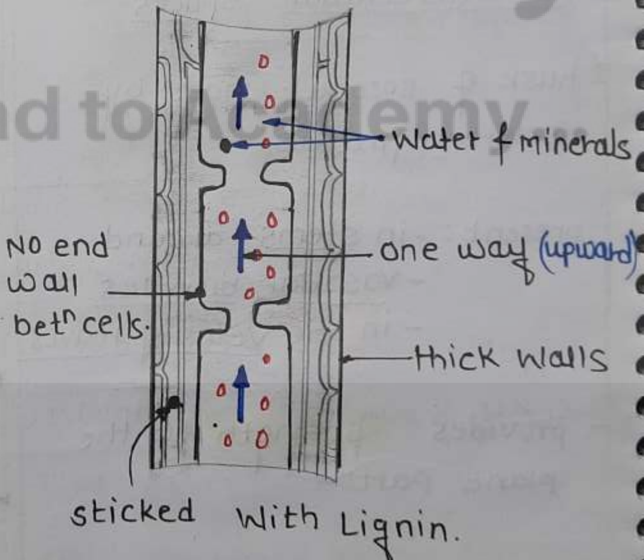
called suberin in theire wall that make them (प्रभल) impervious to gases & water

Complex permanent tissue

- made of more than one type of cells.
- All these cells coordinate to peform a common function.

xylem

- xylem parenchyma – Living cell
- xylem fibre ⟶ dead
- Tracheids ⟶ dead
- vessels ⟶ dead



→ made of dead cell thick cell lining.

Tracheids & vessels:

- they have broad tubular structure for transportation in plants vertically.
- dead cells.

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xylem parenchyma :

- stores Food
- helps in transport of water in plants horizontally.
- living cells.

xylem fibres :

- support in function
- dead cells.

parenchyma are only living cell.

Phloem

- made of living cell
- only phloem fibers is dead cell
- transport of Food form Leaves to
- For movement of food bidirectionally (two way)

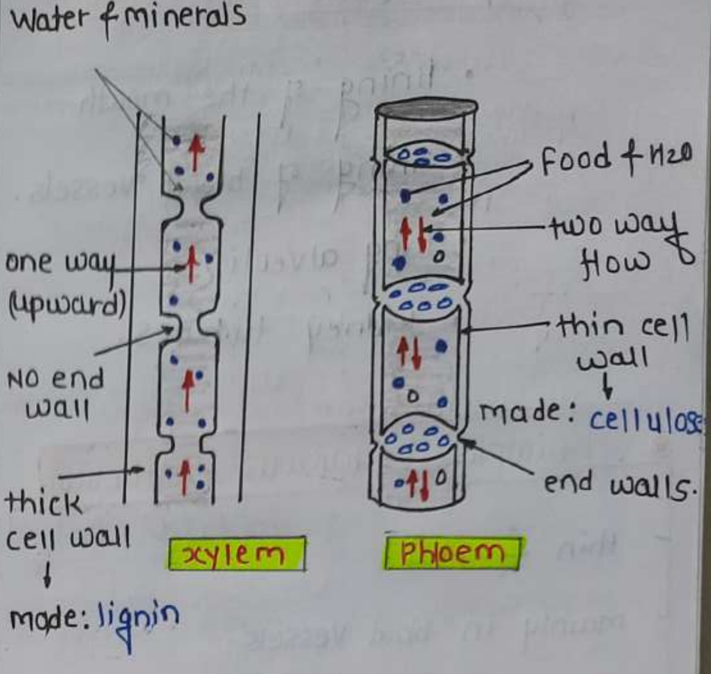
1] **sieve cells** :

2] **sieve tudes** : Transport of Food & Nutrients.

3] **companion cells** : facilitate funⁿ of sieve tubes.

4] **Phloem fibre** : provide flexibility - dead cells.

5] **phloem parenchyma**: stores starch & proteins.



* phloem transports photosynthetic products.

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ANIMAL TISSUES

▸ different : structure, function, origin of tissue.

- ⊙ Epithelial tissue.
- ⊙ connective tissue.
- ⊙ mascular tissue.
- ⊙ Nervous tissue.

Epithelial tissue

- covering or protective tissue
- tightly packed, continuous sheet.
- **Function** : absorption
  secretion
  sensation
  protection,

● Made of E.T : Skin

- lining of the mouth.
- lining of blood vessels.
- lung alveoli.
- kidney tubules.

Simple Squamous epithelium

- thin & flat
- mainly in blood vessels.
- lung alveoli (exchange of gas.)
-

Squamous epithelium

- Found : ~~skin~~
  - oesophagus
  - lining of the mouth

Stratified Squamous epith.

- Found : skin
- arranged in pattern of layers.

Coloumnar epithelium.

- Found : respiratory tract
  - Uterus
  - Fallopion tube
  - central canal of spinal cord.

Cuboidal epithelium

- cube shaped cells
- lining of kidney tubules.
- ducts of salivary glands.
- provid mechanical support

Glandular epithelium

- portion of the epithelium tissue folds inward & multicellular gland is formed.

- Absorption of Nutritients from digested food also takes place through the columnar epithelial.

tall ephith.

inner lining intestine found

cliated colmnar

→ having small hair like "cilia"

Glandular E.

→ sweat, sebum & mucus etc. substances secreted form skin are actually secretions of these glands.

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Connective tissue

- cells are loosely spaced & embedded in an intercelluar matrix.
- matrix may - Jelly like, Fluid, dense or rigid.
- Blood is a type of C.T
- Bone - strong & nonflexible C.T
  cells composed: calcium & phosp.

Blood - has fuild (liquid) matrix called plasma
↓

◉ plasma caintain: proteins
salts
hormones.

Ligament	Tendons
• connect bones with bones	• bones - muscles
	• Fibrous tissue
• very elastic	• limited flexibility.
• considerable strength	• great strength

cartilage :

- widely spaced cell
- composed: proteins & sugar
- present in: nose ear, trachea, larynx.
- smoothens bones surfaces.

◉ cartilage tissue

Adipose tissue

- Fat storing A.T
- Found: below the skin & betn internal organs.
- storage of fats also lets it Act as an insulator.

Areolar tissue

- Found: betn the skin & muscles. As well bone Narrow
- help in repair of tissue.

Muscular tissue

- elongeted cells.
- also called muscle fibers
- responsible for movement in our body.
- contain special protein: contractile proteins.

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Voluntary m.	Involuntary m.
• sketal muscles	• smooth muscle
• striated muscles	• unsriated
• cantrol them	• can't control.
• cells are : long cylindrical un branched multinucleate	cell are : uninucleate long (spindle-shaped)
	* cardiac muscles: -cylindrical - branched - uninucleate

Heart muscle

- cylindrical
- branched
- uninucleate

Nervous tissue

- brain, spinal coard & Nerves are composed of Nervous tissue.
- cell of this tissue are called
  ↓
  Nerve cell or Neurons

→ each Neuron has a single long part called **Axon**.

→ many short, branche part called **dendrites**.

* Nerve impulses Allow us to move our muscles when we want.

synapse : gap betn two Neuron

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DIVERSITY

Whittakar - Five kingdoms classification

- monera
- protista
- Fungi
- plantae
- Animalia

} group formed on the basis of
- source of Nutrition
- body
- mode
- cell structure
- body organisation.

Taxonomy : Identification of living organisms.
- Father: carolus Linnaeus.

Virus : Non - living

* **Algae**: blong to plantae
- perform photosynthesis.
- colour : green / blue / Red

● **carl woese** :

dividing monera
- Archaebacteria (Archaea)
- Eubacteria (Bacteria)

charles Darwin :

→ described idea of evolution.

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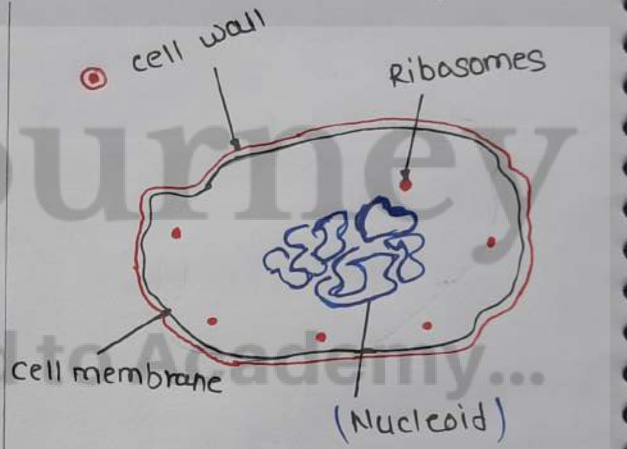
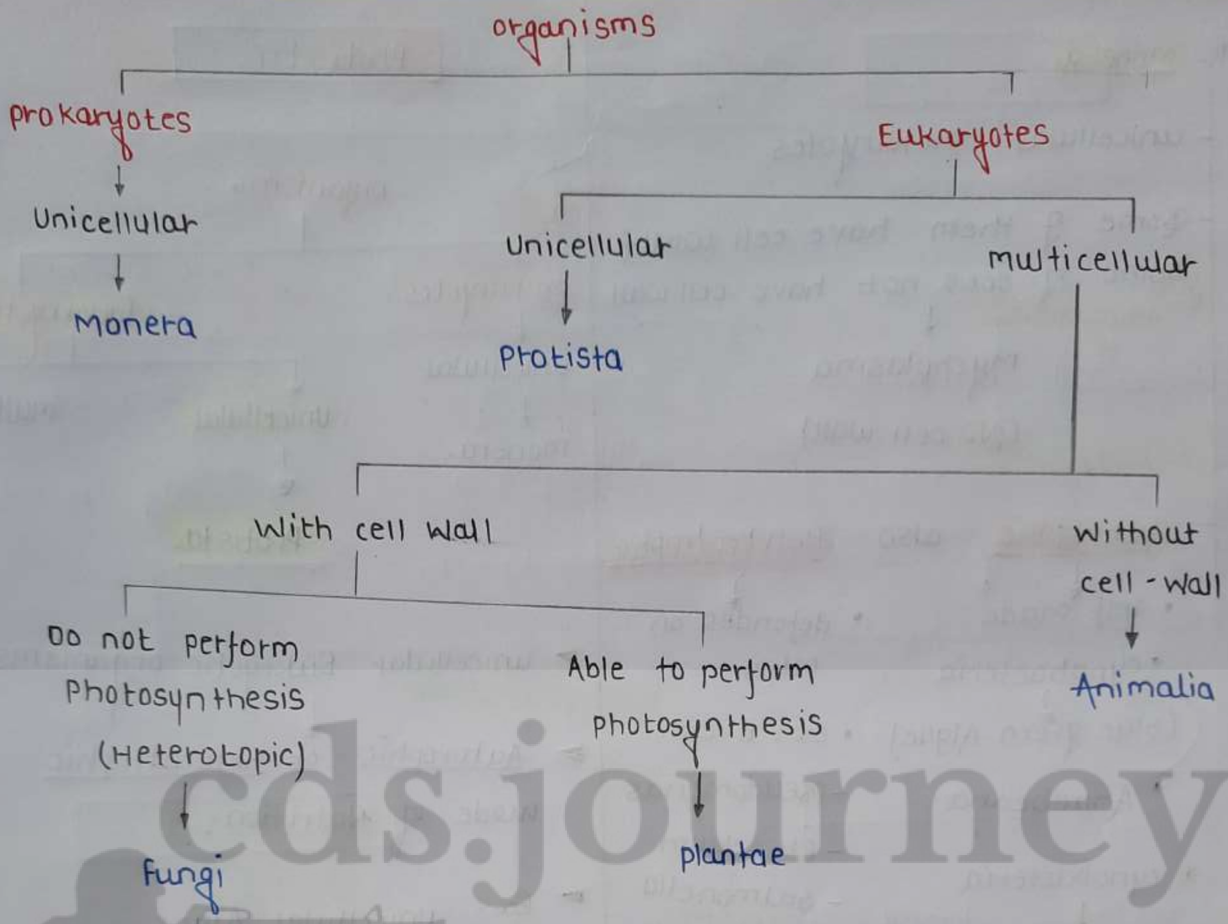


undefind rigion
↓
only in prokary. cell.

Hierachy classification

- kingdom
- phylum - Animals
  Division - plant
- class
- order
- Family
- Genus
- species.



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Monera

* Q. Identify me

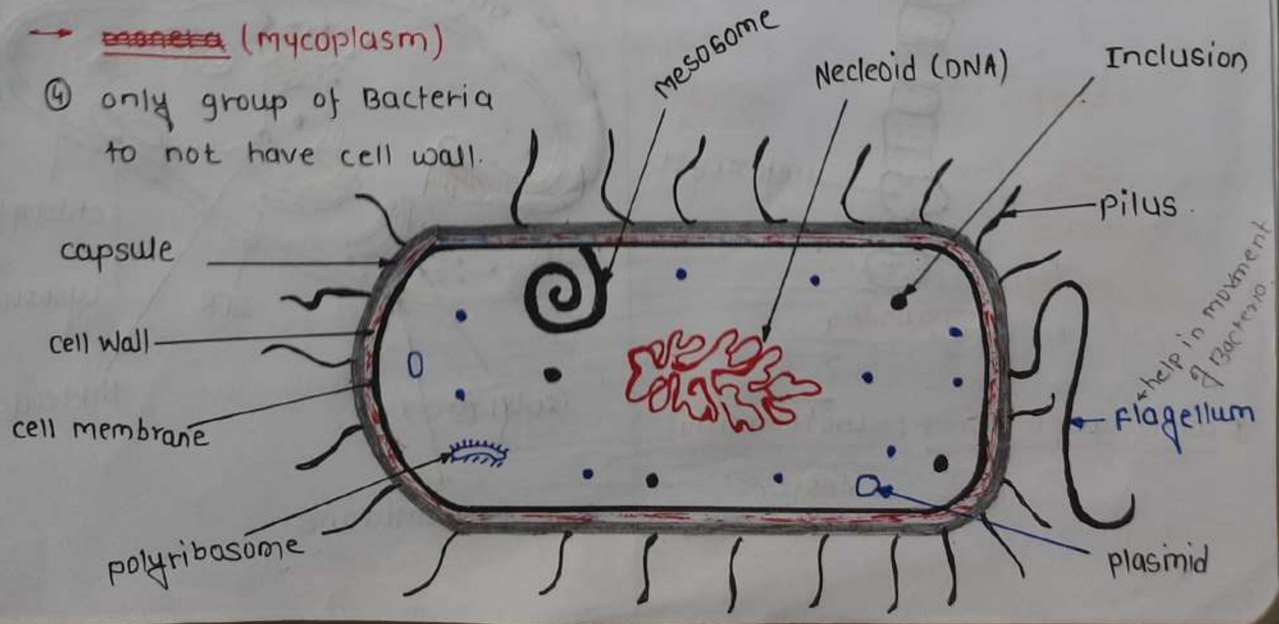
① I have no Lysosomes

② I am the smallest cell

③ come under monera

→ ~~monera~~ (mycoplasm)

④ only group of Bacteria to not have cell wall.

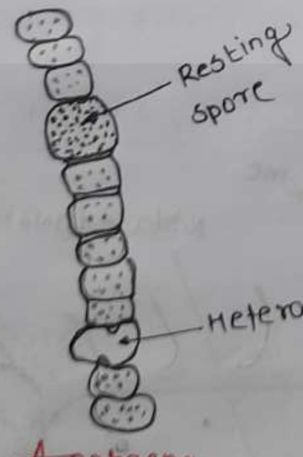


* Monera

- unicellular prokaryotes
- Some of them have cell wall + some of does not have cell wall
  ↓
  mychplasma
  (NO cell wall)
- Autrophic also Hetrtrotrophic

Autrophic	Hetrtrotrophic
• self made	• defendes on other.
• cynobacteria (blue green Algae)	• Ex: E. coli
• Anaebeana.	- pseudomonas - Rhizobium - salmonella - Agrobacterium

* cynobacteria
↓
chlorophyl (present)
↓
photosynthesis.



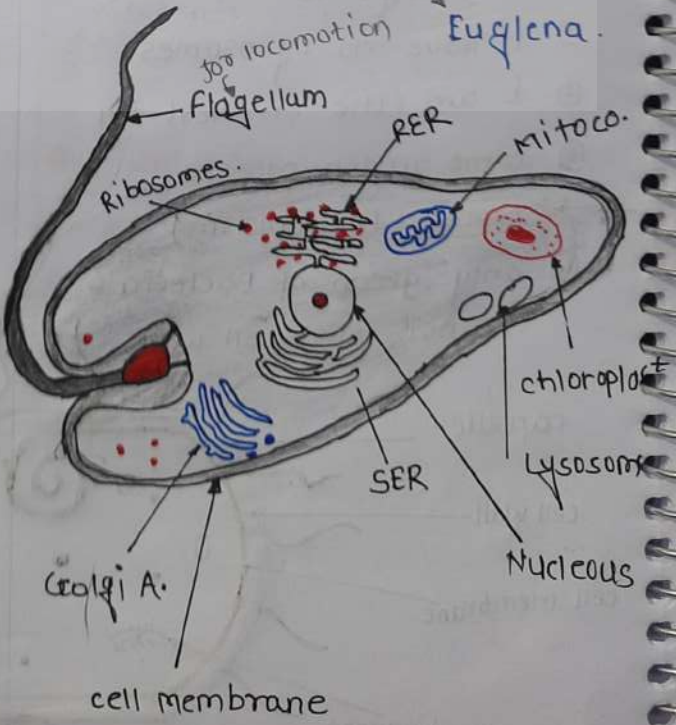
Anabaena

* come under moner: Lactobacillus
Nostoc.

Protista

organisms
- prokaryotes → unicellular → monera.
- Eukaryotes
  - Unicellular → Protista
  - multi-

- unicellular Eukryotic organisms.
- Autorophic or heterotrophic mode of Nutrition.
- Ex. unicellular Algae
  diatoms
  protozoans → paramecium, Amoeba, Euglena.



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fungi

organisms
- prokayotes → Unicellular → Monera
- Eukaryotes
  - Unicellular → protista
  - Multicellular
    - With cell wall
      - Do not perform photosyntesis → Fungi → cell wall made - chitin (though complex sugar)
        - Heterotrophic eukaryotic (Depends on others)
          - parasite | require a living protoplasm of a host organism for food
          - Saprophytic mode of Nutrition in Fungi | Feed on dead
        - unable to perform photosynthesis
          - Ex: yeast
          - (bread)- molds (bread)
          - mushroom
          - Aspegillus
      - ✓ → plantae
    - Without cell | Animalia


(yeast)

- Fungi - Break down the food outside the body.

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Plantae

Eukaryotes

↓

multicellular

↓

celleoulos ← with cell wall

↓

Able to perform photosyntesis ← use chlorophyll

↓

mode of Nutrition: Autotrophs

Cryptogams

(Non seed producing plant)

- Thallophyta
- Bryophyta
- pteridophyta

Phanerogams

(seed producing plant)

- Gymneosperms
- Angiosperms.

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Thallophyta

- Not have well differentiated body.
- vascular tissue (xylem, phloem) - Absent
- seeds - Absent
- Flower | fruit - Absent
- Habitat.: Aquatic plant
- plant of this group called - Algae
- Example:
  - Spirogyra
  - Ulothrix
  - cladophora
  - Ulva
  - Chara
  - Lichen

Bryophyta

- called Amphibians of the plant kingdom.
- vascular tissue - Absent
- seeds | Flower - Absent
- Fruit - Absent
- differentiated to form
  - ↓ steam
  - ↓ leaf lite structures
- Example:
  - moses(Funaria)
  - marchantia
  - Riccia.

∴ no true roots

Pteridophyta

- Body design - Differentiated
  - roots
  - steam
  - leaves
- vascular tissue - present
- seeds / Flower - Adsent
- Fruit - Adsent
- Habitat - Terestrial
- They consist of the embryo Along with stored food.
- Example: ⊙ marsilea
  - ⊙ fern
  - ⊙ horse tails.

* seedless fruit

- orange
- pineapple
- Fig
- Grapes.
- Bananas

Phanerogams

Angiosperm

- Flowering plants
- Enclosed seeds
- Flower - present
- Fruit - present
- Ex: ⊙ Dicot →

- ⊙ monocot →

- Plantae
- vascular t. present
- seed present

Gymnosperm

- Non - Flowering plant
- Naked - seeds
- No Flower
- No Fruit
- Ex: • cycas
  - pines (Pinus)
  - deodar • cedrus

Thallophyta	Bryophyta	Petridophyta.
• No body differentiation	• body differ.	• body differ.
• Aquatic	• Amphabiam	• Terestrial
• No vascular tissue	• NO v. t	• yes - vascalar tissue present.
• seeds/ Fruit / Flower - ✗	• ✗	• Fruit - present
• Ex: Algae spirogyra	• mosses marchantia Riccia	• marsilea fern

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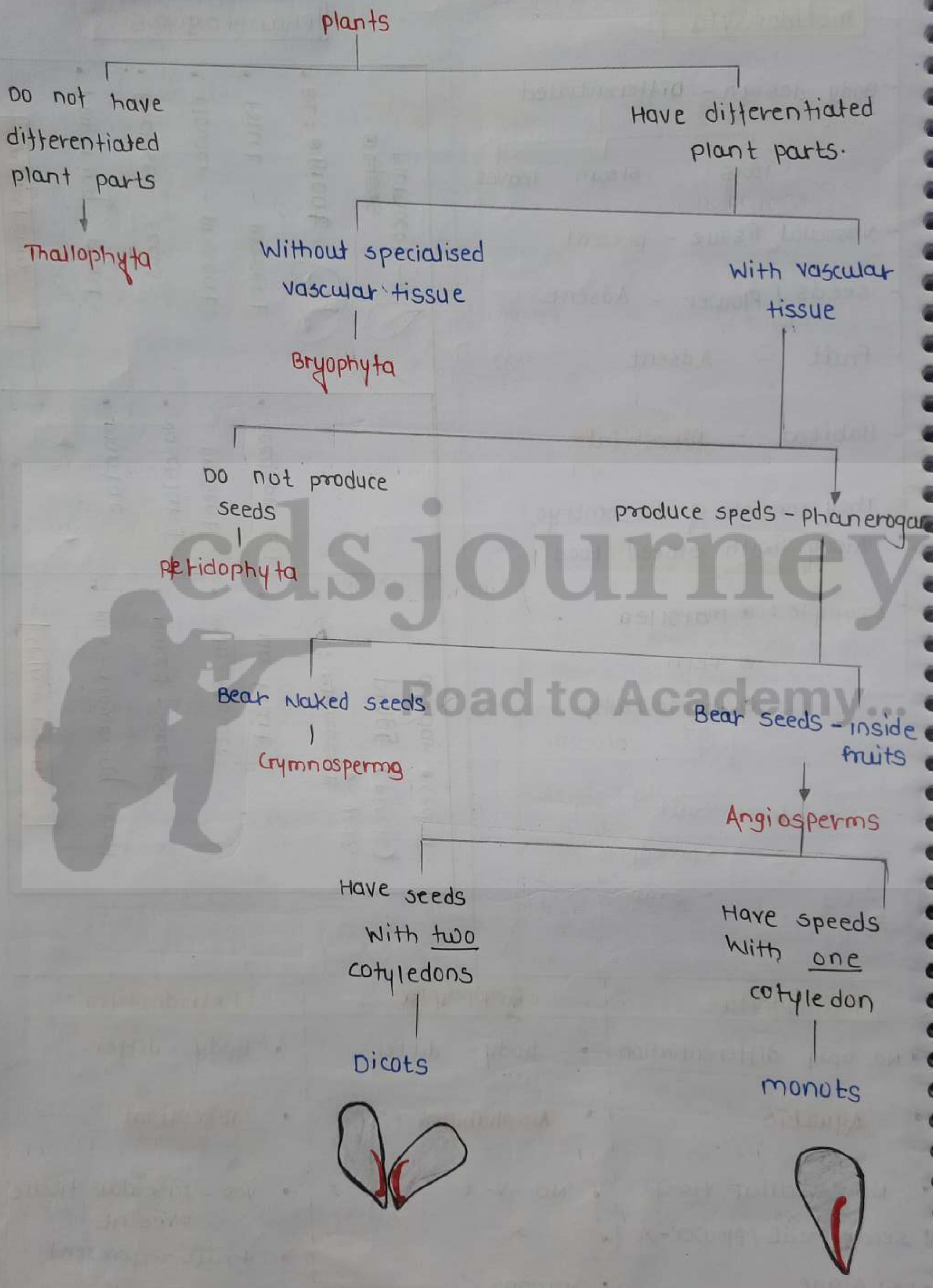

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Animalia

Eukaryotes

multicellular

Without cell-wall

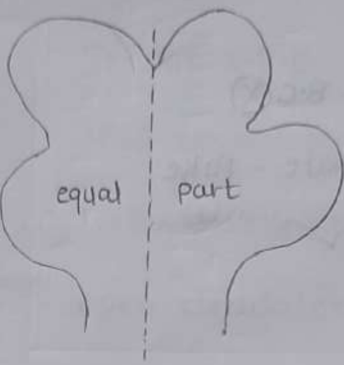
mode of Nutrition: Heterotrophic

Symmetry

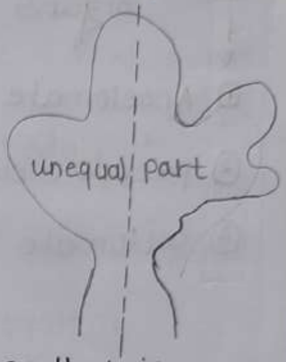
- Porifera - Asymmetry
- colenterata - Radial sym } Diploblastic ②
- platyhelminthes - Bilateral S
- Nematoda - B.S
- Annelida - B.S
- Arthropoda - B.S
- mollusca - B.S
- Echinodermata - Radial sym.
- chordata - B.S

} ③ Triploblastic

* symmetry:

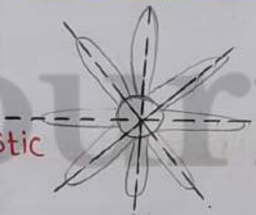


* Asymmetric



can't divide in equal part

* Radial symmetry



starfish

- any vertical plane equal.

* Bilateral sym.

- Divid.only Left & Right

Butterfly

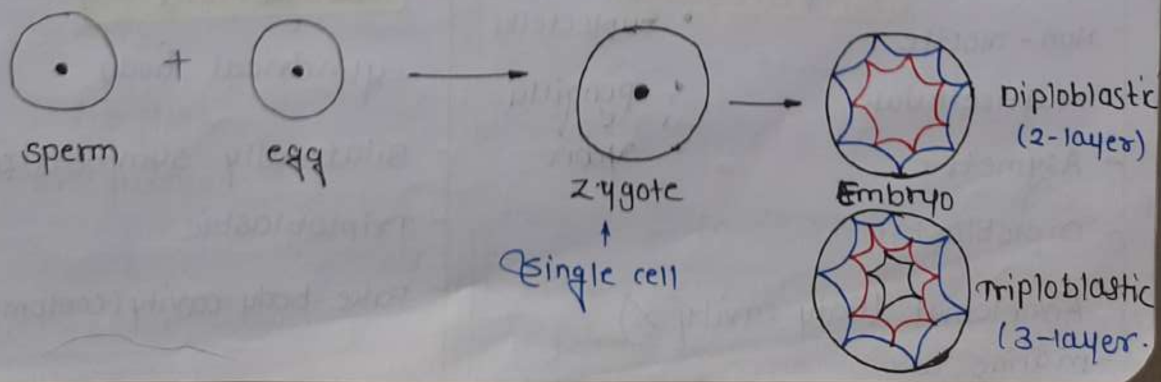
* Germinal layers

↓

Embryonic layers

- Diploblastic
- Triploblastic

*



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Body cavity (coelom):

↓

arrange internal organs

① Acoelomate - B·C ⊗

② pseudo coelomate - Fake

③ coelomate - ✓

- porifera
- coelenterata
- plathyhelminthes

} Acoelomate

- Nematoda Fake → peseudo coelomal
- Annelida
- Arthropoda
- mollusca
- Echinodermata
- chordata

} coelomate

porifera (pores)

- with hole - body not differentiated.
- known as sponges
- Non-motile
- multicellular
- Asymetric
- Diploblastic
- Acoelomat (Body cavity ×)
- marine also.

Ex:

- Euplectella
- spongilla
- sycon.

coelenterata (cnidaria)

- body is differentiated into 2 ends
- all aquatic Animals
- two layers of cell (Diploblastic)
- Radial symetry
- Acoelomate B·C (×)
- Ex: Hydra
  Jellyfish
  Anemones

plathyhelminthes

- known as Flatworms
- bilaterally symmetrical
- Triploblastic
- NO true internal body cavity or coelm
- Ex: platyhelminthes
  Tapeworm
  planaria - free living.
  Liver fluke - parasitic

Nematoda

- cylindrical body
- Bilaterally symmetrical
- Triploblastic
- Fake body cavity (coelom)

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Ex: Nematoda
Ascaris
Wuchereria.

Annelida

- Segmented cylindrical body
- body differentiated into head + tail.
- Bilaterally symmetrical
- Triploblastic
- Have true body cavity
- Ex: Earthworm, Leech.

Arthopoda

- largest phylum in the Animal
- Jointed legs
- bilaterally symmetrical
- we differentiate organ (B.C) ✓
- Triploblastic
- open circulatory system.
- Ex: spiders
  butterflies
  mosquitos

Mollusca

- kidney like organ → excreatry sy.
- Bilaterally semmetrical
- Triploblastic
- less segmented body
- Well developed organ (B.C) ✓
- open circulatory system
- Limbas are present
- Ex: snails + octopus

Echinodermata

- Radial symmetry + Triploblastic
- true coelom cavity
- Have Hard calcium carbonate skeleton. structure
- Free living marine Animals.
- Ex: sea urchins
  starfish.

Chordata (Proto)

- bilaterally semmetrical
- Triploblastic
- circulatory system is closed type.
- Road like system
- Marine Animals

Ex - Balanoglossus
Herdmania
Amphixus.

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Vertabrata

- have a notachord
- vertebral column
- internal skeleton
- Bilaterally symmetrical
- Triploblastic . coelomic & segmented

cyclostomata

- body is round & elongated like an eel.
- skin soft & smooth
- ② chambered heart
- lateral line acts as a sense organ.
- Ex: petromyzon, maxine

Pisces

- body is usually streamlined.
- cold blooded organisms
- gills help in respiration.
- lays - eggs
- closed type of single blood circulation.
- ② - chambered heart
- ~~Sharke~~ - both bone & cartilage.
  ↑
  tuna, rohu.
- sharke - made entirely cartilage
- Dog fish - scoliodon.
- Lion Fish - Pterois Volitans
- Flying fish - Exocoetus

Amphibia

- lays egg
- ③ chamberd
- both land & water
- respire through the lungs & skin. either Gills
- Gills present
- Eg: Frogs, salamanders 3 chambers
- mucus glands in the skin.

Reptilia

- clod - blooded
- breathe through lungs.
- ③ chamberd heart
  ↓
  only crocodiles - ④ chambere.
- skin dry, rough, without glands.
- Ex: snakes, Turtles, Lizards. chameleon.

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Aves (birds)

- warm bboded
- 4-chambered heart
- lay eggs.
- breathe through lungs.

MAmmalia

- warm blooded, external ears.
- 4-chamber heart
- have hair, oil glands ← sweat.
- platypus & echidna (exception)
  ↓
  lays eggs.
  ↓
  as well milk

* Notochord - long rod like structure

cold blooded	Warm blooded
- cannot maintain constant temp.	- maintain temp.

closed circulation	open circulation
- blood flow inside vessels ↓ arteries, veins	- No vessels. - No blood capillaries.

vs

Q. In vertebrates ⟶ SER cell of Liver
↓
plays a crucial role in
Liver ← detoxification of toxic compounds.

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Life - processes

Nutrition

- Autotrophic
  - chemo-Autotrophic
    - ↓ using chemicals
    - • few bacteria
    - - ~~chemoo~~
  - photo Autotrophic
    - required
    - ↓ CO_2 + sunlight + water
    - • plant
- Heterotrophic
  - Holozoic N.
    - engulf
    - digest inside body
    - • Human
    - • Amoeba (Animal)
  - Saprotrophic N.
    - depends on dead & decay organisum.
    - ↓
    - • Fungi
    - • mashroom
    - • mould
    - ↓ Food break down outside.
  - parasitic
    - without ~~living~~ kiling & oraganism
    - • cuscuta
    - • ticks, lice
    - ↓ • Tapuoams

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Autotrophic Nutrition

► CO_2 + water $\xrightarrow[\text{chlorophyll}]{\text{sunlight}}$ carbohydrate & $(C_6H_{12}O_6)$ + O_2

Glocoge.

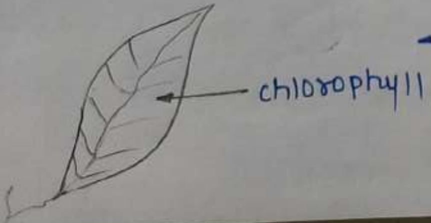
H_2O → $\boxed{H_2}$ + O — Split

↓ plant which is not used immediately stored in form of Starch.

• Animals ↓ Food is stored in the form of ↓ Glycogen

Reduction

► $6CO_2 + 12H_2O \xrightarrow[\text{chlorophyll}]{\text{sunlight}} C_6H_{12}O_6 + 6O_2 + 6H_2O$



← ① light energy absorbed by chlorophyll.

② Light energy converted by chemical energy.

* during the process of photo-synthesis.

- ◉ Absorption of light energy by chlorophyll.
- ◉ conversion of light enery to chemical energy. & splitting of H_2O molecules into H_2 & O_2
- ◉ Reduction of CO_2 to carbohydrates.

* exchange of gases in plant through.

- leaves
- stems
- roots

* desert plants

↓

take up CO_2 → At Night

* Some cells contain green dots.

↓

These green dots are cell organelles

↓

called: chloroplasts

↓

cantain chlorophyll

↓

essential for photosynthesis.

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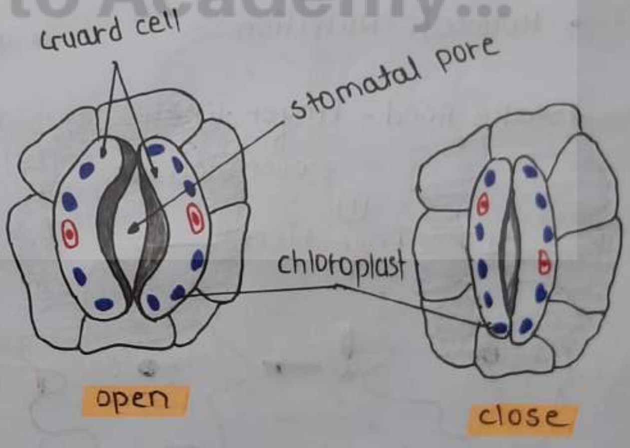
Stomata

- gaseous exchange take place in the leaves.
- large Amout of water lost -
- present on the surface of leaves.
- close stomatal pore - Not want to CO_2.
- guard cells: Fun- opening & closing of the pore.

swell ↓ (when H_2O flow into them)

shrink ↓ pores close.



- Nitrogen, phosphorus, iron & magnesium are take from soil by Autotrophs.

- Nitrogen - essential for synthesis of proteins.

Heterotrophic Nutrition

Holozoic N.

- engulf Food
- Food break down inside the body

- Animals
- Amoeba

saprotrophic

- depends on dead & decay organisum.
- Food break down outside the body.
  - yeast
  - mushrooms.
  - bread moulds.
  - Fungi

parasitic

derive Nutrition from plants & Animals without killing them.

- cuscuta (Amar-bel)
- ticks
- lice
- leeches
- tap-worms

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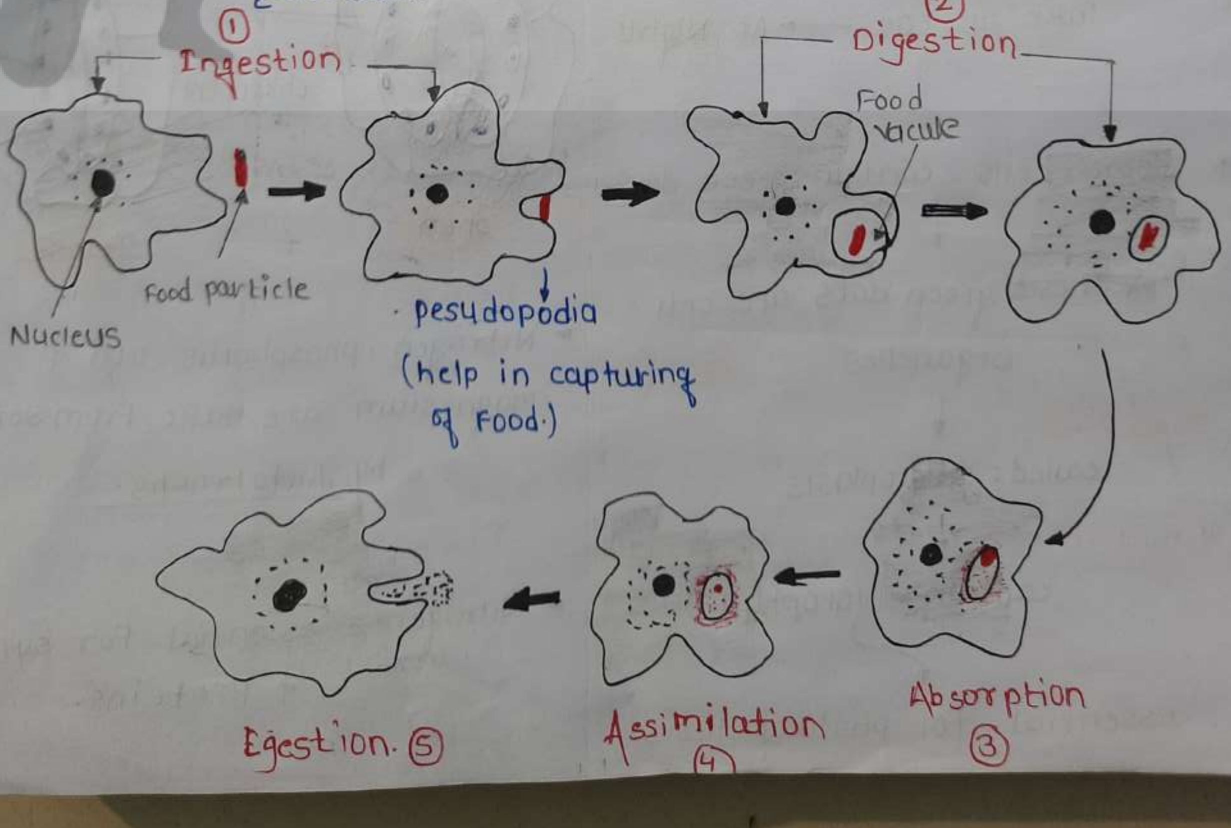
Amoeba

⊙ Ingestion → Digestion → Absorption → Assimilation → Egestion.

- Holozoic Nutrition
- take Food - Finger like extension.

* paramoecium → cilia



Nutrition in Human Beings

- Digestion start from mouth

↓

Fluid called saliva in mouth

↓

saliva contains enzyme called salivary Amylase.

↓

complex starch (maltose) Break down into simple sugar.

→ Food : mouth to stomach through Oesophagus (Food pipe)

* Alimetary canal

↓

- Organs thats Involve to pass Food.
- mouth
↓
oesophagus
↓
stoma
↓
small
↓
large

\# musc

help o
mixin

* Accessory organs

↓

- help in digestion of food
- Liver
→ Bile
juice
Pancreas

\# the small intestine also produce its own enzymes.

- peptidases
- sucrase
- Lactase
- maltase

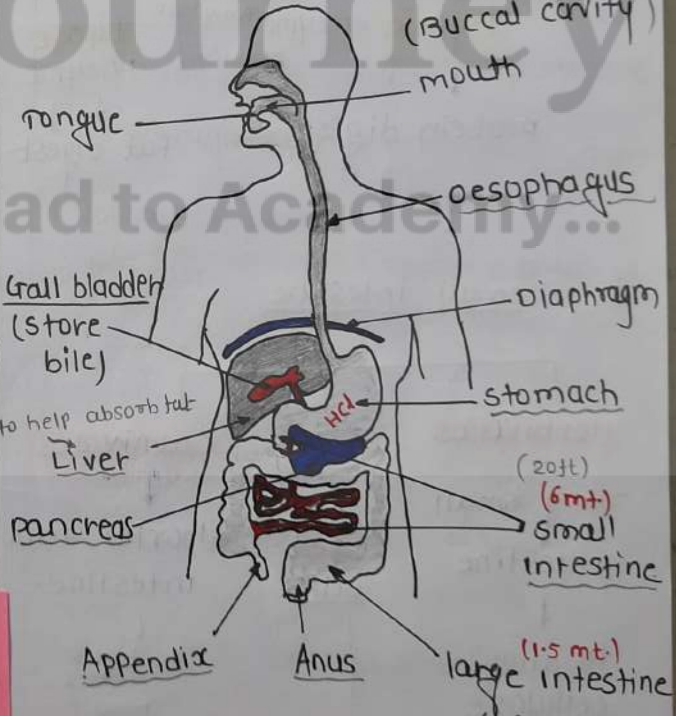
(sucrase, Lactase, maltase) contribute mainly to the hydrolysis of Polysaccharid

◉ gastric glands :

Fun: digestion in stomach is taken care.

present : in wall of the stomach

pepsin	mucus
• protein digesting enzyme • incomplet digestion in stomuch.	• protects the inner lining of the stomach.



* stomach — HCl, pepsin, mucus

↓

incomplet digestion

↓

small intestine

complet digestion

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*sphichter muscle :
↓
regulate Food from the stomach.

* Gall bladder
↓
bill juice → PH : 8

Food (acidic)

pancreas
↓
Pancratic Juice → PH - 8.5

basic

small instestine ← •longest par of Alimentry canal.

(complete digestion)

fat, protein, chardohyd.

Trypsin enzyme
↓
protein digest

Amylase

Lipase enzyme
↓
Fat digest.

small intestine

Herbivores
- long small intestine
↓
to allow cellulose to digest

villi
↓
enzymes finall convert.
↓

Carnivores
↓
shorter small intestine.
↓
ex tiger → meat

- protein → Amino Acid
- complex carbohydrates → glucose
- Fatty Acid → glycerol

small intestine
↓
- Duodenum: region which follows the stomach
↓
- Jejunum : middle part
↓
- Ileum : later region
↓
- large intestine

Respiration

Breathing	Respiration
• process - exhaling + inhaling ↓	• Break down ↓ glucose ↓ + produce energy
• olny lungs	• all cells
• physical process	• chemical process
• No energy	• ATP produce.

Respiration

Aerobic	Anaerobic
• O_2 required	• O_2 Not required or lack.
• take place cytoplasm + mitco.	• cytoplasm only (yest)
• complet breakdown of glucoce.	• Incomplet break.. of glucose
• End process : CO_2 + H_2O	• End prod - CO_2 + Alcohol, Lactic Acid.

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Glucose (6-carbon) $\xrightarrow{\text{In cytoplasm}}$ pyruvate (3-carbon)

- Absence of O_2 (in yest) (Anaerobic R.) → Ethanol + CO_2 + Energy (2-carbon)
- Lack of O_2 (muscle cell) → Lactic acid + energy (3-carbon)
- presence of O_2 (mitochondria) ↑ (Aerobic resp.) → CO_2 + H_2O + Energy

- release of energy

Aerobic process > Anaerobic process.

- Lactic Acid : cause cramps.

- plant : exchange of gases through stomata.
  ↓
  human - lungs
  ↓
  Alveoli

- rate of breathing

Aquatic organism $\overset{\text{Faster}}{>}$ terrestrial organisms.

- In human : air taken through
  ↓
  Nostrils (in Nose)
  ↓
  Nasal passage
  ↓
  Nasal cavity
  ↓
  pharynx
  ↓
  Larynx
  ↓
  Trachea
  ↓
  Bronch → Lungs → Bronchioles
  ↑
  Alveolar Sacs
  ↑
  Blood capillary

Inhalation	Exhalation
• Ribs lift up	• Ribs downward
• Diaphragm flat	• Diaphragm dome shaped. ⌒
• volume of lungs increase	• volume of lungs decrease.

- Haemoglobin present in the RBC.
  ↑ carry O_2 (RBC)
  ↑ very high affinity for oxygen. (Haemoglobin)

- If the alveolar surface were spread out cover - about $80 m^2$

- CO_2 is more soluble in water than O_2.

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Transportation

plants
↓
Transportation
↓
done by: vascular tissue.

humans
↓
Transportation
↓
circulatory system.

Blood → carry everything also (salt)

plasma
55% of blood
↓
fluid medium called plasma.
↓
content:
- protein
- Hormones

Blood cells (45%)

Sycle cell Amenia

RBC
↓
Shape:
↓
circular + discshaped
red colour becose
↓
Himoglobin
↓
comman Name: erythrocytes.

WBC
large + colourless (Leucocyte)

Platelet
↓
help
↓
blood clotting

㊉ **cockroach** → 13 chamber heart

㊉ **octopus** → 3 heart

Arteries	**Veins**
• Flow of oxygenated blood	• Flow of deoxygenated blood.
• Located: deep in the muscle.	• veins present surface of the body.
• Have very thick wall.	• Have thin walls.
• blow flow: heart → body	• blood carry: toward heart.

◉ **coronary Artery:**
↓
carries blood to heart muscle

◉ **capillaries**
↓
conection b/w arteries + veins

◉ **carotid Artery:**
↓
supplies oxygenated blood brain to eyes.

◉ **subclavian Artery**
↓
providing blood
↓
head + Neck.

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⊕ Arteries - have thick, elastic walls

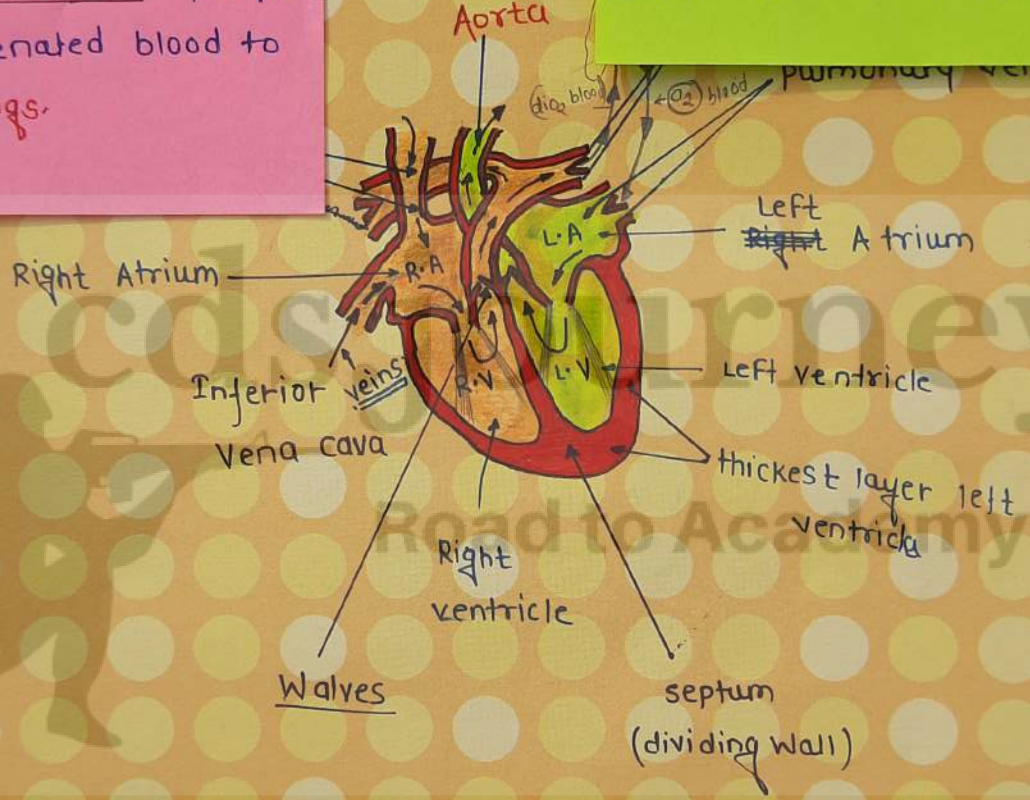
⊕ veins - not Need thick walls.

Right Atrium - receives deoxygenated blood from the body & pums it to the right ventricle.

Right ventricle: pumpthe deoxygenated blood to the lungs.

Left Atrium: receives oxygen-rich blood from the lungs and pumps it to the left ventricle.

left ventricle: left ventricle pumps the oxygen rich blood to the body.



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⊕ veins - carry deoxygenated Blood towards heart

⊕ Arteries - carry oxygenated blood away from heart.

⊕ pulmonary Arteries - only Arterie can carry deoxygenated blood
↓
heart to lungs

⊕ pulmonary veins - only veins can carry oxygenated blood
↓
lungs to heart

⊕

blood pressure

↓

Force that blood exerts against the wall of a vessel.

↓

BP - Arteries > veins

systolic pressure

↓

P. of blood inside artery during ventricular systole (contraction).

↓

Normal S.P (about 120 mm of Hg)

diastolic pressure

↓

pressure in Artery during ventricular diastole (relaxation)

↓

Normal D.P (80 mm of Hg)

Lymph

↓

colourless & contains less protein.

↓

absorbed fat from Intestine & drains excess fluid from extra cellular space back into the blood.

Nitrogenous Waste

① urea
② uric acid
③ CO_2
④ Excess water.

Excreation

excretory stem of human beings.

↓

pair of kidneys

↓

pair of ureters

↓

Urinary bladder

↓

Urethra.

Kidney → PH maintain

↓

located : in the Abdomen

↓

urine produce in the kidney

↓

Ureters

↓

Urinary bladder: urine stored

↓

Urethra

each kidney has 1 million Nephron.

↓

Filtration unit of our body.

◉ Bowman's capsule: cup-shaped end of a coiled tube.

◎ osmotic pressure ↑ - cause H_2O to move tusse.

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Guttation	Transpiration
● Night	● Day
● water loss in the form of water droplets	● loss of water in the form of water vapour.

* O2 - Wast in plant

* plant waste ⟶ stored in
↓
cellular vacuoles

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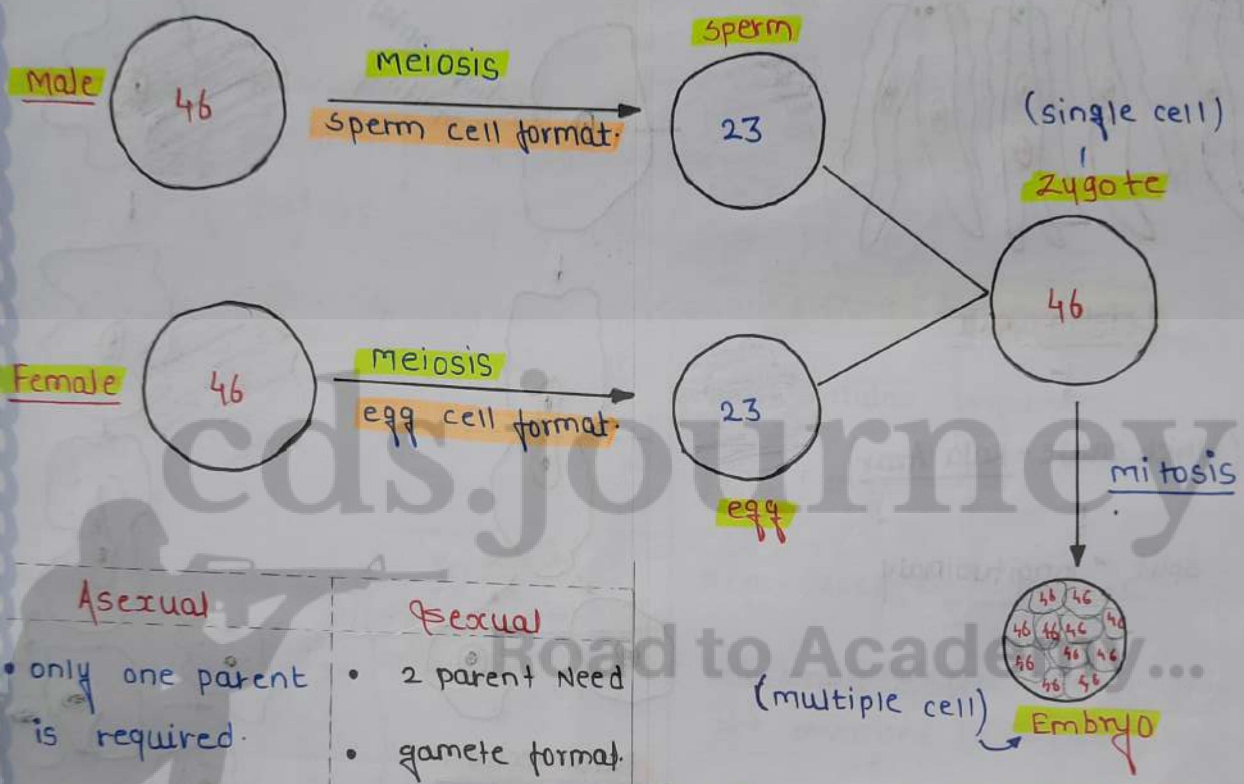
REPRODUTION

- **chromosomes** : becomes half during gamete formation.



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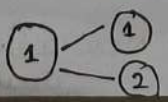
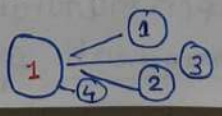
Asexual	Sexual
only one parent is required.	2 parent Need
No gamete	gamete format.
No fertilizer	fertilization
No zygote	zygote

Fission : spliting (or) division

binary
↓
organism split into two part.



Multiple
↓
organism split into more than two part.



⊙ The DNA in the cell Nucleus is the information source for making proteins.

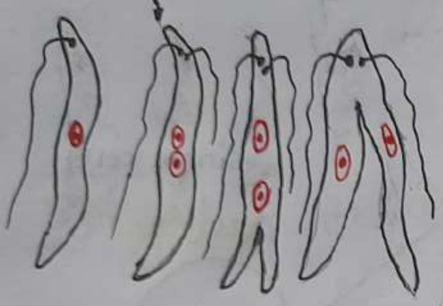
⊙ basic event in reproduction is
↓
creation of a DNA copy.
↓
DNA copies

Will be similar
↓
but may not be Identical.

Some variation each time
↓
useful for survival of species over time

Binary Fission

Longitudinal binary fission



Leishmania

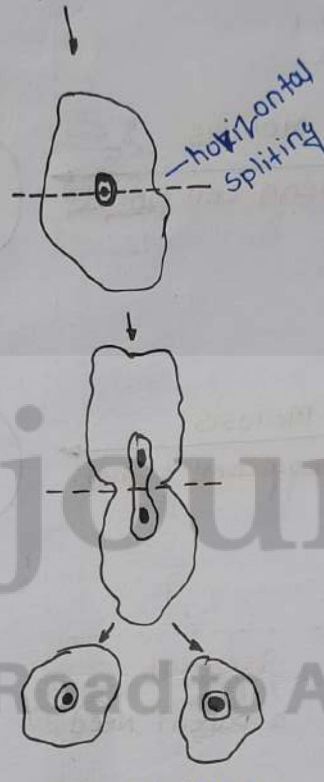
↓

(Which cause - kala Azar)

↓

split - longitudinaly

Transverse binary fission



paramecium

Irregular binary fission.

Amoeba

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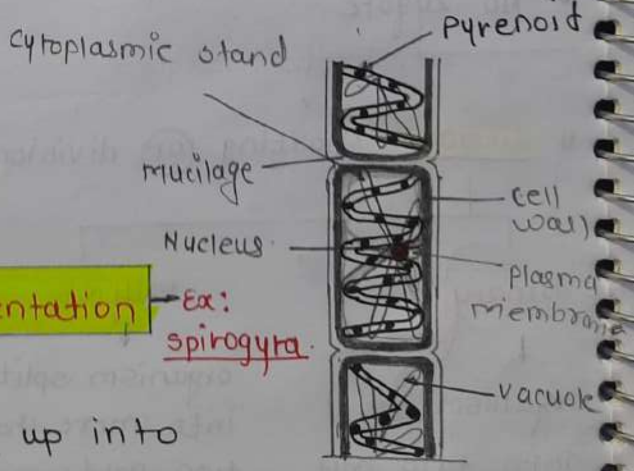
Multiple Fission

↓

- malarial
- parasite
- [redacted] plasmodium



Fragmentation

→ Ex: spirogyra.

← multi-cellular organisms.

↑

but Not ture for all mco.

↓

simply breaks up into smaller pieces upon maturation. these pices or Fragment grow into New Idividuals.

Budding

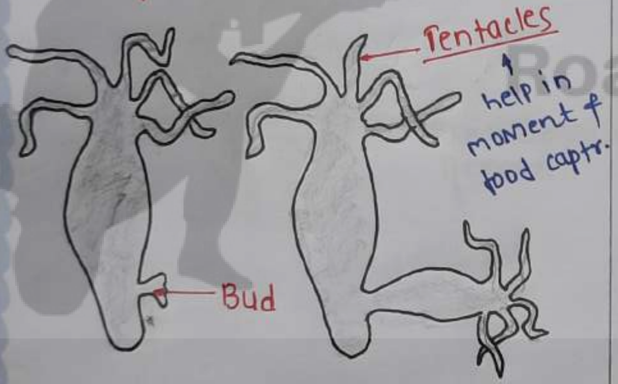
bud develops as an outgrowth due to repeated cell division at one specific site.

These buds develop into tiny individuals & when fully mature.

Example

Hydra

yeast



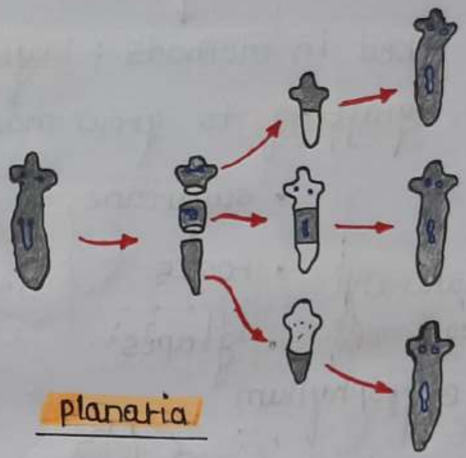
Regeneration

individual is somehow cut or broken up into many pieces, many of these pieces grow into seperate individuals.

Example ⟶ lizard

Planeria

Hydra.



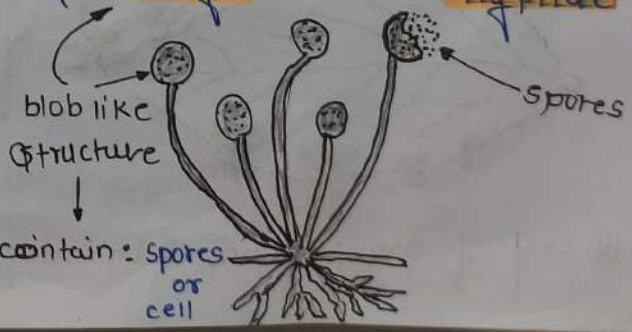
planaria

Spore formation

- multicellular organisms
- Fall: Fungi
- Reproduces by spores
- blob-on-a stick structures are involved in reproduction
  ↓
  called: sporangia
- Ex: Rhizopus

Reproductive part
↓
Sporangia

Non-reproductive part
↓
Hyphae



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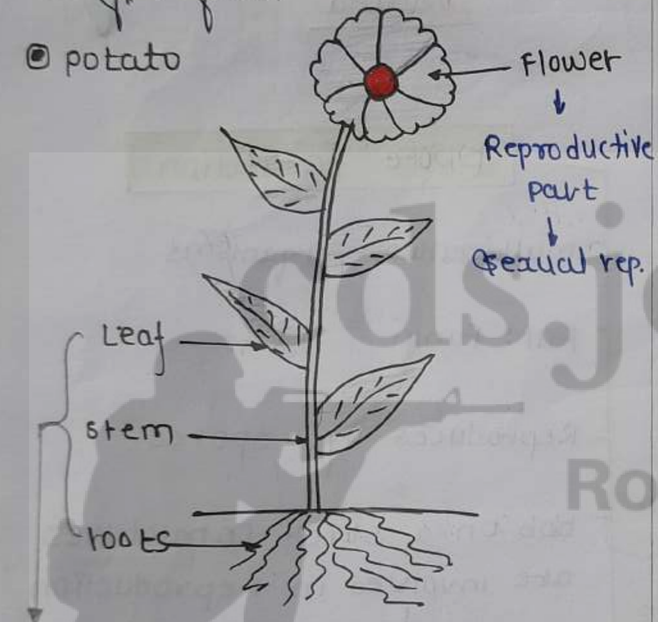
vegitative propagation

used in methods : layering (or) grafting to grow many plant.

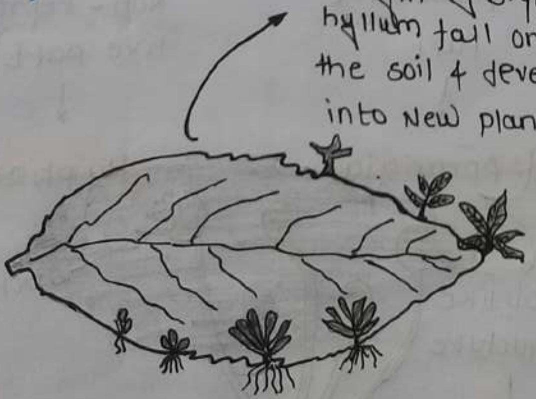
- sugarcane
- roses
- grapes

} Artificial veg. prop.

- Natural veg. prop.
- Brypophyllum
- potato



*bud produced in the Notches along the leaf margin of Bryophyllum fall on the soil & develop into New plant



Leaf of Bryophyllum with buds.

Natural vegetative propagation

- leaves - bryophyllum
- stem - turmeric, ginger
- runners / stolon - grass runners, strawberry.
- bulbs - onion, lily.

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Sexual Reproduction

Flower

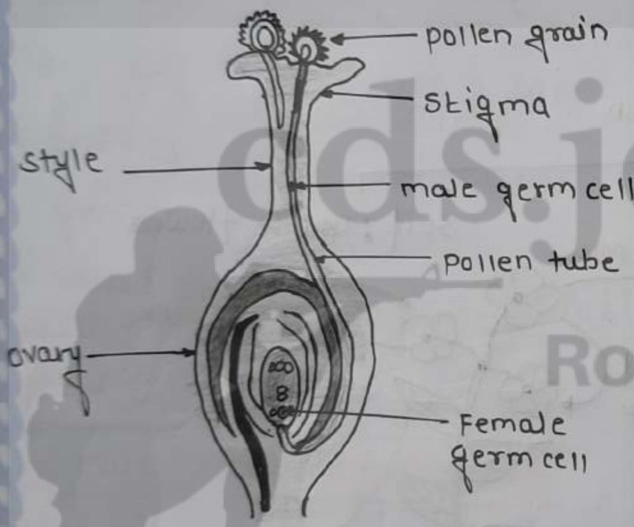
Female part — reproductice part — **mel part**

[Pistil | carpel]

- stigma
- style
- ovary

[Stamen]

- Anther
- filament



Anther
↓
pollen grain (produced by male germ cell)
↓
male gamet.

⊙ Stamen
↓
produce pollen grains
↓
that yellowish in colour

Flower

unisexual

- stamenate ↓ male Reprodu-etive part
- pistilate ↓ female repro-ductive part

ether one

Ex: papaya
Watermelon

Bisexual
↓
contains both stamens + pistil
(male) (femele)

Ex:
- Hibiscus
- mustard
- Pea.

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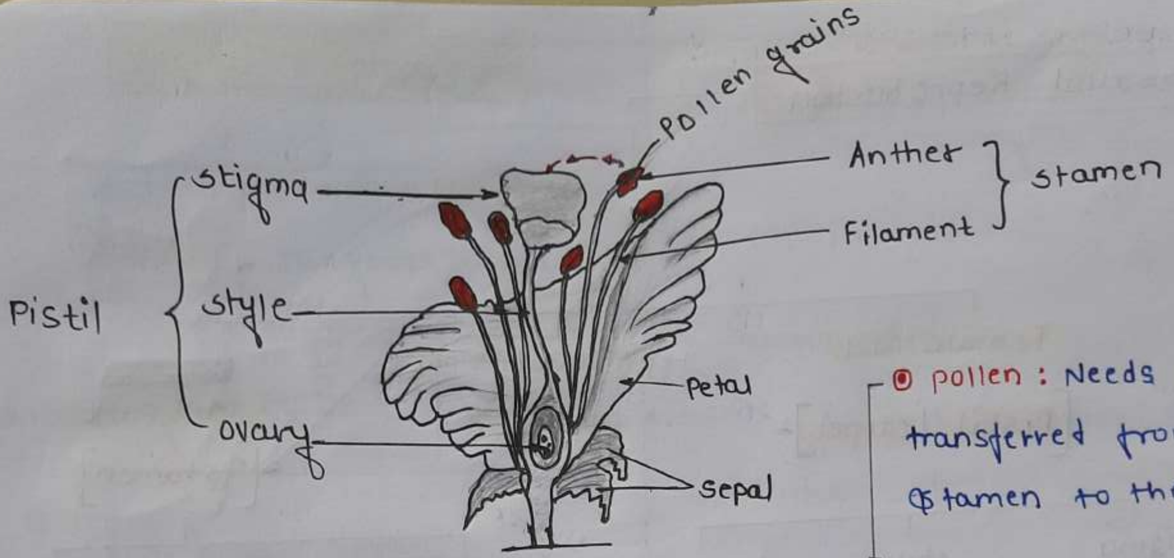


- pollen : Needs to be transferred from the stamen to the stigma.

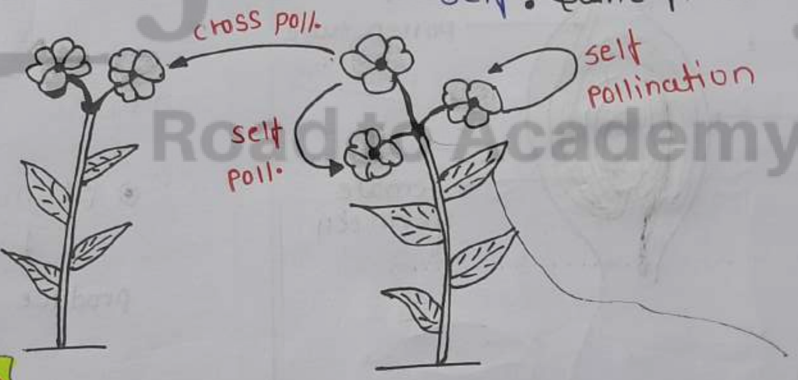
↓

produced in <u>Anther</u>

pollination

Cross (one flower to another)
↓
through • Wind
• Animals
• Water

self : same flower



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After fertilization

↓

- egg converted → zygote : single cell

↓ ↓

embryo : multiple cell

↓

seed ← ovule converted

↓

fruit : ovary grows rapidly & ripens to form fruit.

- Germination

↓

cotyledom (Food store)

Sexual Reproduction in Human

puberty

boys	girls
Facial hair growth	hair : underarm + pubic
thinner hair also in legs, arm	Breast develop.
skin oily	darkening skin of Nipples at tips of breasts
genitals growth	mensuration cycle.
voices deepning	voice shrilled (patle)

Male Reproductive system

- testes : formation of sperms (or) germ cells.

↓

- located outside the abdominal cavity in scrotum (pouch).

↓

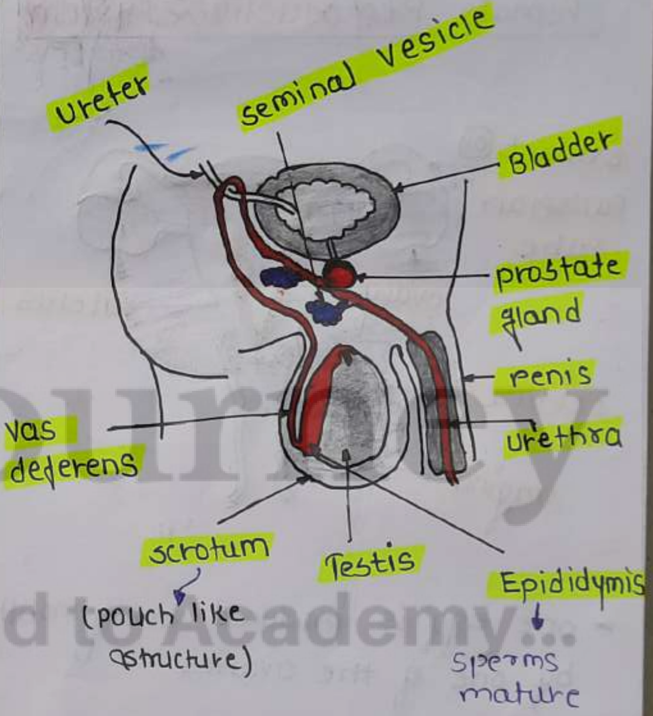
sperm formation requires a lower temp. than normal boy temp.

◉ sperms : are produced by seminiferous tubules and mature in the epididymis

* Leydig cell (or) interstitial cells.

↓

present : b/w the seminiferous tubules secrete hormone testosteron



◉ Vas-deferens : transports mature sperm to the Urethra.

◉ Urethra : common passage for both the sperms and urine.

◉ Along the path of the vas deferens

↓

in sperm

↓

prostate glands & seminal vesicles

↓

add their secretion (Nutrition)

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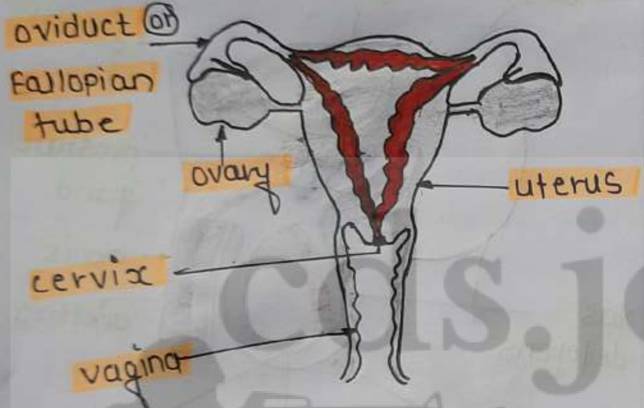
↓
so that the sperms are Now Fluid
↓
make their thaansport easier.

Female Reproductive System



- one egg is produced every month by one of the ovaries
- egg is carry form ovary to the womb through a thin oviduct.

◉ **uterus:** two oviducts units into an elastic bag-like structure.

◉ sperm + egg > zygote → embryo

zygote: single cell (Fertilized egg)

embryo: divide by multiple cell

- foetus

◉ **placenta:** The embryo gets Nutrition from the mother's blood with the help of a special tissue.

- villi
↓
provide
↓
large surface area for glucose and O_2 to pass from the mother to the embryo.

- The child is born as a result of rhythmic contractions of the muscles in the uterus.

◉ main reproductive organ - pairs of ovaries.

◉ produce female sex hormones
→ estrogen
→ progesterone

◉ female ovary
↓
produce egg once a moths
↓
uterus wall become thick

egg fertilized
↓
pregnancy

Not fertilized by sperm
↓
Egg + uterus wall + blood vessels
↓
Rupture → come out through the vagina → blood & mucous

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Control & Coordination

▸ Nervous system

CNS [central Nervous system]
- Brain
- spinal cord

PNS [Phipheral Nervous system]
- spinal Nerve
- cranial -II-

▸ Gustatory receptors : detect taste

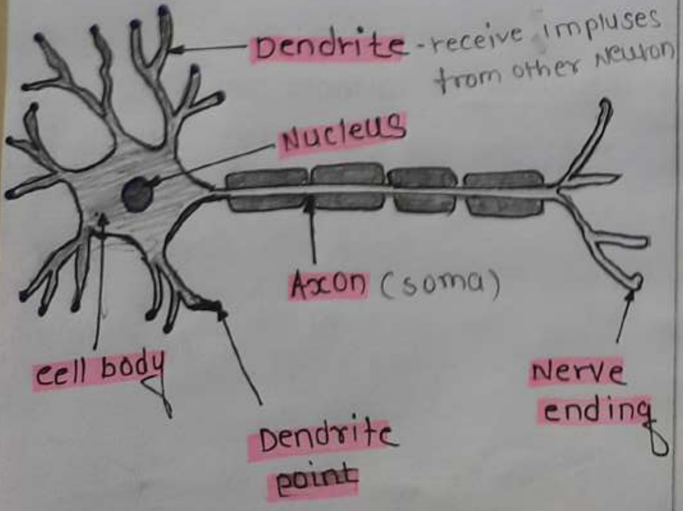
▸ olfactory receptors - smell

▸ photoreceptors - Eyes

▸ Thermo receptors - skin

▸ Sphenoreceptors - Ears

▸ Inner ear - Body balence.



Neuron

◉ synapse: gap between two Neuron.

↓

b/w two Neuron

↓

Neurotransmiter

◉ Types of Neurons

1] sensory Neuron

2] Motor Neuron

3] Interneuron / Relay Neuron

* Sensory Neuron: [Afferent]

→ carry information receptor to Brain.

* motor Neuron: [Efferent]

→ from brain cells to effector organ.

* Interneuron

→ sensory Neuron

↓

motor Neuron.

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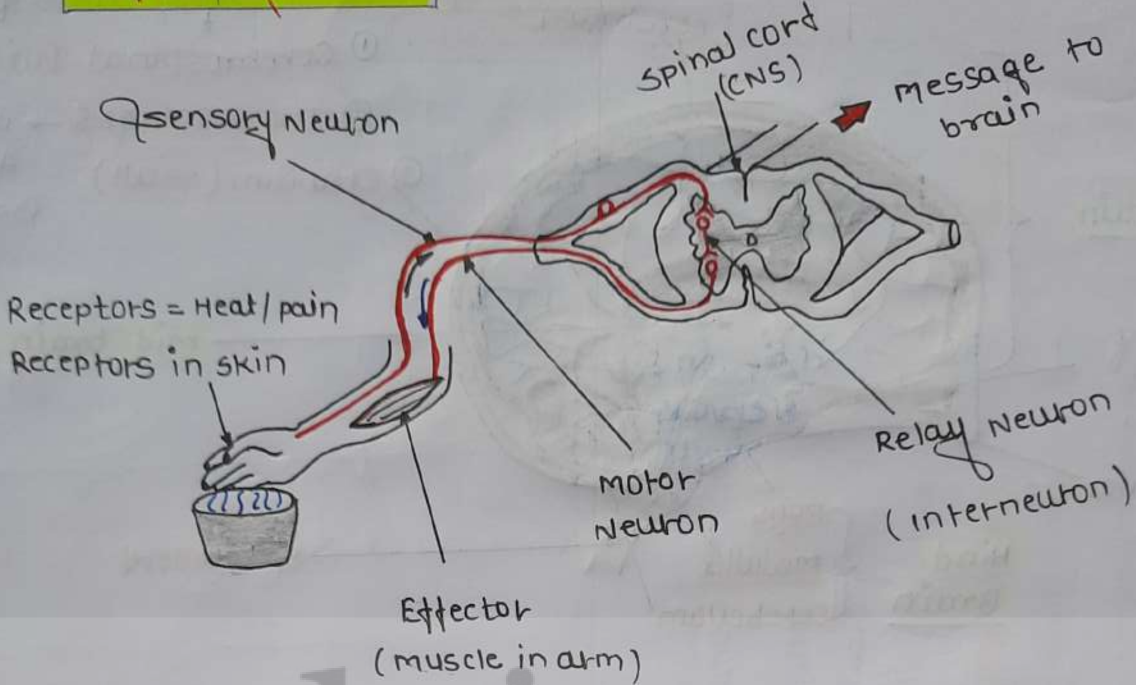
* Refex Action



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⊙ stimulus receive

↓

receptor

↓

sensory Neuron

↓

Relay Neuron (in spinal cord)

↓

motor Neuron

↓

effector

↓

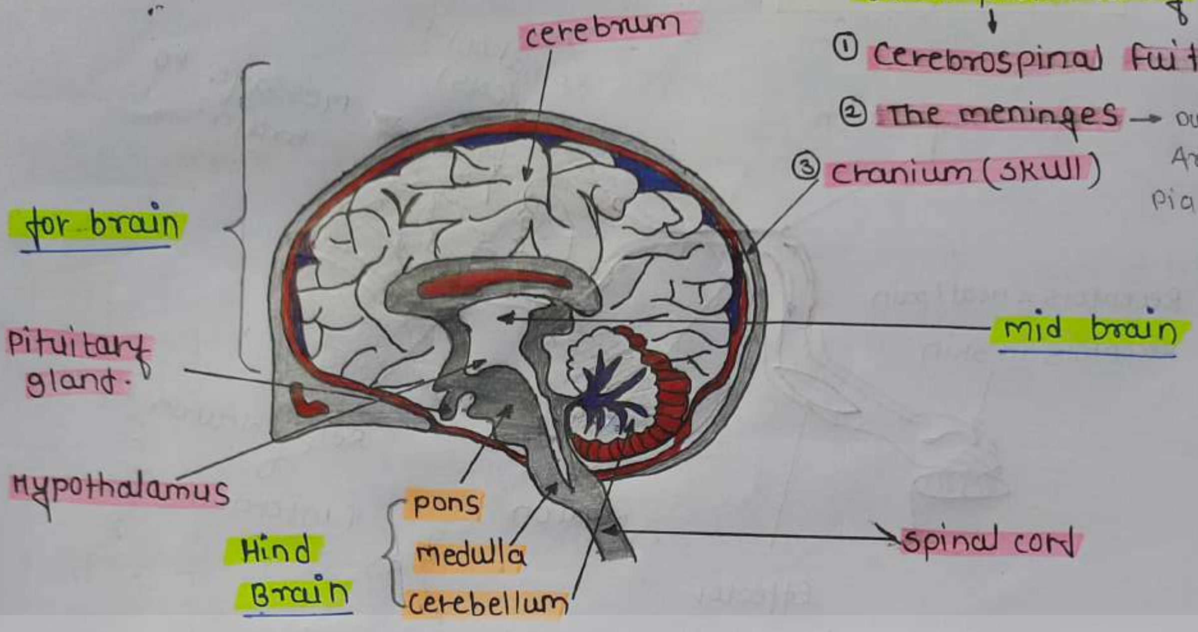
response

* Brain is not involved in Refex Action $\boxed{\text{But}}$ info. will reach Brain.

* vertebral column (or) backbone

↓

protects the spinal cord.



brain protected by
↓
① Cerebrospinal fluid
② The meninges → Dura mater
Arachnoid
Pia mater
③ Cranium (Skull)

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Brain

for-brain

- receive singnles form sensory organ.
  - eyes, ears, skin nose, tongue.
- voluntary movment
  ↓
  Past experience.
- fore brain is the main thinking part of the brain.
  ↓
  regulate the sense of
  - sight
  - smell
  - hearing.

mid brain

↓

connect for brain & hind brain

↓

controlled: involuntary Action.

Hind Brain

↓

→ ⊙ cerebellum:

↓

Balance of body

- walk in a straigh line
- riding bicycle
- picking up pencil.

→ ⊙ medulla

↓

- control involuntary action
  - Blood pressure
  - slivation
  - vomiting
  - Heart Beat
  - cough
  - respi. rate.

← ⊙ pons

- sleep cycle
- Breathing rate.

Plant Hormones

Plant Hormones

- Growth promoters
  - Auxin
    - help in growth longer in plant tissue.
    - light coming one side
      ↓
      Auxin diffuses towards the shady side of the shoot.
    - plant to bend towards the light.
  - Gibberellin
    - help in growth of
      - stem
      - germination
      - flowering
  - Cytokinins
    - help in cell division
    - Delay ageing
- Growth Inhabitors
  - Abscisic Acid
    - Inhibits growth.
    - wiltering of leaves.
    - Dormancy of buds./seeds.
- Growth inhabitors + promoters.
  ↓
  - Ethylene
    ↓
    this is gaseous hormmone which causes
    ↓
    ripening of fruits

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Hormones in Animals

hormones are secreted by
↓
endocrine glands.

↓
If contraction of the diaphragm & rib muscles
↓
breathing rate increase.

Gland

Thyroid Gland
↓
Iodine is nacessary to make thyroxin Hormone.
↓
Idione: syntesis of thyroxin
↓
Idione deficient:
↓
goitre disease
symptoms:
↓
swollen Neck
↓
Thyroid Gland
↓
Regulates metabolism for body growth.

pituitary
↓
stimulates growth in all organs
↓
deficiency: Hor. in childhood
↓
- It leads to dwarfism

Pineal gland
regulate sleeping Pattern

pancreas
Insuline
help in regulate blood sugar level.
only Gland
endocrine
exocrine
secreat insuline
↓
diabetes: taking injections of insuline.

adrenal Gland
↓
regulete:
- heart beat
-

testes
Sperms

ovaries
ovum

* 12-10 years (puberty)

male
secretion of testosterone

Female
secretion of oestrogen.

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Growth independent movements

↓

Nastic movements

movements which are not growth related.

↓

thigmonastic movement

↓

movement in response to touch

↓

Ex: touch-me-not plant

↓

belongs to: mimosa family.

Growth-related movements in plants.

↓

tropic movmemnts

↓

growth related.

Ex: phototropism: response in light

(+ve) pho.

↓

growth towards the light

- stem
- soot

-ve photo

growth away form the light

- root

Geotropism

In responds to gravity

(+ve) Geotropism

↓

- roots

(-ve) Geo.

- stem

Thigmotropism

↓

in respose to touch & an object.

23

pineal Gland

↓

pituitary Gland

↓

Thyroid gland

↓

Thymus

↓

Adrenal Gland

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Heredity and Evolution

* **Genetics** : study of heredity & variation.

↓

father of Genetics - G.J mendel

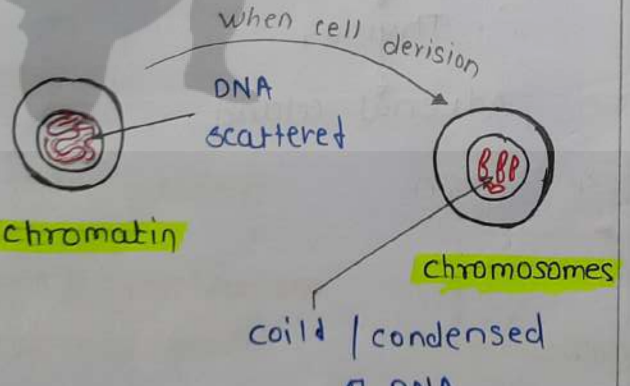
↓

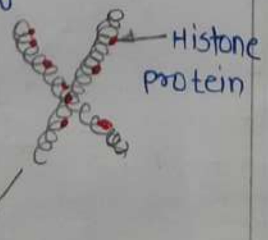
worked on pea plant

* traits

somatic variation	gametic variation
(Acquired traits)	(Inherited)
↓	↓
(after birth)	(by birth)
• tattoo	• Hair colour
• Driving	• Eye colour
	• skin colour.

*





* Gene

↓

A functional segment of DNA

Eg:

*

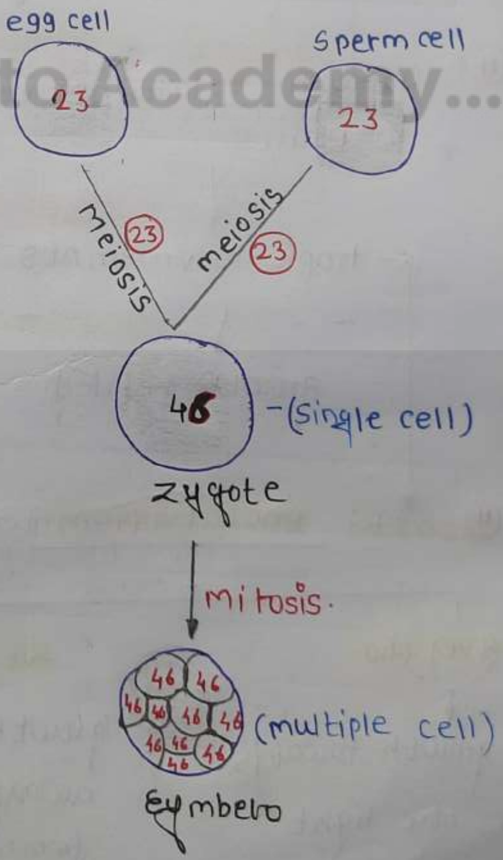
Haploid | Diploid

→ All human cell are diploid

Except gamets

Sperm	egg
Haploid	Haploid

*



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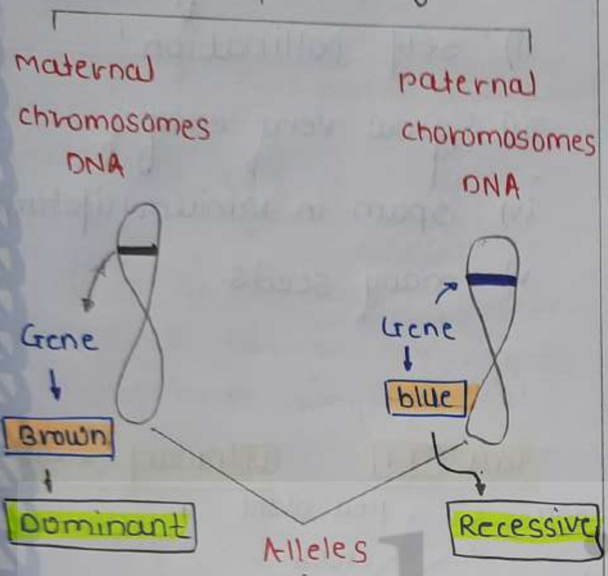
* **Alleles**

↓

Gene ki variety



↓

Stronger gene will win.

↓

(Dominant)

↓

Brown

* Dominant Allele	Recessive Allele
↓	↓
denoted by capital letter Ⓣ	denoted by small letter ⓣ

* pea → Pisum sativum

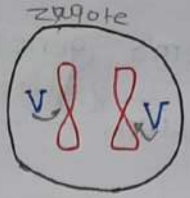
Flower

- violet flower (V) - Dominant
- white flower (v) - Recessive.

\#

i)

zygote



⟹ Violet colour

⇓

Homozygous dominent condition (or) (Pure condition)

VV

↓

both are Dominant

ii)

zygote

⟹ white colour

⇓

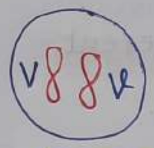
Homozygous Recessive condition (or) (pure condition)

vv

↓

both are Recessive

iii)



⟹ Violet colour

⇓

Heterozygous condition (or) Hybrid

Vv

one Dominant + one Recessive

↓

Next generation will be Dominant

- vv - White - Recessive
- Vv - Violet - Dominant
- VV - violet - Dominant

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Genotype: organism's gene's information.

- VV ⇒ dominant homozygous
- Vv ⇒ heterozgous
- vv ⇒ homozygous (Reccissve)

phenotypes: Physical traits

purple   white

eye से दिखेगा

Homozygous - Identicle

Heterozygous - Different.

	Dominant	Reccessiv
seed		
shape	Round	Wrinkled
colour	yellow	Green
flower colour	violet	white
pod shape	Full	contricted
pod colour	Green	Yellow
flower position	Axial	Terminal
stem length	Tall	Dwarf.

Why pea plant taken?

i) many varietys

ii) self pollination

iii) grow very easyly.

iv) spam in Minium lifetime.

v) many seeds.

Tall TT   tt Dwarf (small)

pollens   pea plant   ovum

cross

Monohybrid

F1 - progeny (Generation)

	t	t
T	Tt	Tt
T	Tt	Tt

← Tall

→ All plant are Tall

→ Tall Hetrozygous condition

F2 - Self pollination

Tt × Tt

	T	t
T	TT	Tt
t	tT	tt

- Tall - 3
- Dwarf - 1

Tall	pure Tall	small
2	1	1

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- ratio of Tall & Dwarf

3:1

- ratio of pure tall & Dwarf

1:1

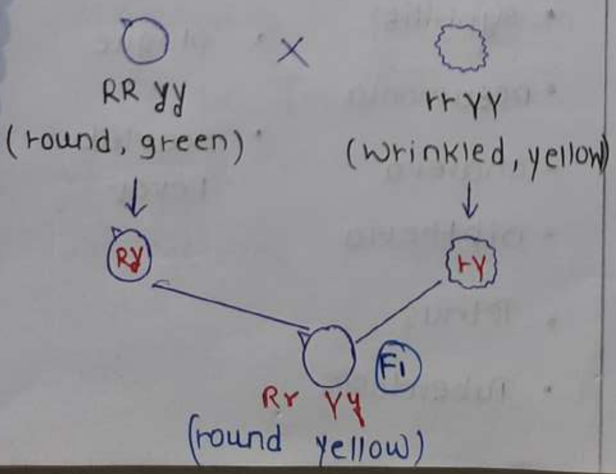
- What is the ratio of Heterzygous tall to Homozygous dwarf in F_2 progeny?

2:1

✱ Phenotypic Ratio of $F_2 \rightarrow$ Dihybrid.

* Round yellow - 9
* Round Green - 3
* Wrinkled yellow - 3
* Wrinkled Green - 1

F_1



Father XY mother XX

	X	Y
X	XX	XY
X	XX	XY

← Genotypic

- XX - girl
- XY - boy

} phenotypic

homologous

- Common origin but different fun^n
  - looks same
  - fun^n: different

Analogous

- different origion, common fun^n.
  - looks - different
  - fun^n - same.

Food chain

- 4th tropic level - large carniva. & tertia carni.
- 3rd tropic level - small carniva.
- 2nd tropic level - herbiva.
- 1st tropic level - Autotraps.

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Diseases

* congenital diseases

↓

- present since birth
- Imbalance of chromosomes.

* Aqquird diseases

Communicable Diseases
- Bacterial
- virus

Non-comm. Diseases
- heart Attack
- Diabetic

* Deficiency (Diseases)

children body growth stop

- Kwashiokar - protein
- marasmus - protein
- goiter - Iodine
- Anaemia - Iron
- Hypokalemia - potassium
  (Heart beat fast)
- rickets - vitamin D
- scurvy - vitamin C

Diseases caused by Viruses

↓

Antibiotics are not usefull against virus.

↓

use vaccine

- cold
- Flu
- Smallpox
- Measles
- mumps
- AIDS
- Ebola
- chicken pox
- Rabies
- Hepatities
- polio
- Dengue
- mers

Diseases caused by Bacteria

- cholera
- Gonorrhea
- syphilis
- pneumonia
- chalera
- Diptheria
- Tetnus
- Tuberclosis
- Leprosy
- pertussis
- plague
- scarlet fever.

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Disease caused by protozoa

- Kala-Azar – Leishmania
- sleeping sickness – Trypanosoma
- malaria
- Amoebic Dysentry
- Toxoplasmosis

1) Helicobacter Pylori – peptic ulcers
2) Leishmania (proto) – Kala-azar
3) staphylococci – Acne
4) Trypanosoma (proto) – sleeping sickness

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epidemic diseases

i) cholera
ii) smallpox
iii) malaria – plasmodium
iv) yellow fever
v) polio
vi) Measles.

⊙ Down's syndromes - 47 chromoso.

(#) Disease

Endemic ↓ small region

pendemic ↓ World
- covid